



A Study of Duality and Optimality in Bilevel Multiobjective Programming Problem Using Convexifactors



Dissertation

Submitted in Partial Fulfilment of the Requirements for the Degree of

Bachelor of Science
In Mathematics Honours

Submitted By

Yashi Singh

22033563011

Semester-VIII

Under the supervision of

Mr. Hari Kishan Bhardwaj

Department of Mathematics

Kalindi College

University of Delhi

2025–2026



Certificate by Supervisors and TIC



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Under the Supervision of		Teacher-In-Charge
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2025–2026



Self-Declaration for Originality of Work

I, **Yashi Singh**, hereby declare that the work presented in this dissertation entitled "*A Study of Duality and Optimality in Bilevel Multiobjective Programming Using Convexifactors*" is my original work, carried out under the guidance of my supervisor, Mr. Hari Kishan Bhardwaj. No part of this has been stolen, copied, duplicated or taken; all sources of information used in this work have been duly acknowledged through proper citations. I take full responsibility for the content of this dissertation and declare to be the rightful and only owner of all the work done in this dissertation.

Date: _____

Signature: _____

Place: New Delhi

Name: Yashi Singh



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List of Abbreviations

BLPP	Bilevel Programming Problem
CF	Convexifactor
CQ	Constraint Qualification
D	Wolfe Type Dual Problem
D1	Wolfe Type Dual (scalar bilevel)
D2	Mond Weir-Type Dual (scalar bilevel)
KKT	Karush–Kuhn–Tucker (Conditions)
LCF	Lower Convexifactor
MFCQ	Mangasarian–Fromovitz Constraint Qualification
NACQ	Nonsmooth Abadie Constraint Qualification
P	Bilevel Multiobjective Optimisation Problem
RBLPP	Reformulated Bilevel Programming Problem
RP	Reformulated Single-Level Problem
RP₁	Unconstrained Auxiliary Vector Problem
UCF	Upper Convexifactor
VOP	Vector Optimisation Problem

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Abstract

This dissertation prepares a rigorous and unified theoretical structure for this study in bilevel multiobjective programming using convexifactors. Bilevel programming is used to model hierarchical decision problems where lower level's solution set derives the feasible region of the upper-level optimisation problem. When both levels involve vector-valued objectives, or when the data functions are nonsmooth, classical smooth methods and Clarke-subdifferential methods become inadequate. Convexifactors—closed sets that encode directional derivative information without requiring convexity or compactness—provide a sharper and more flexible analytical tool in this setting.

The dissertation combines and builds upon four original articles. The convexifactor calculus of Dutta and Chandra, including chain rules, a mean-value theorem, and a strict hierarchy of lower and upper semi-regular convexifactors, is presented as an analytical foundation of the study. Their optimality conditions for weak efficient points in vector optimisation are formulated through separation arguments and support-function representations. These ideas are then extended, following Suneja and Kohli, to scalar bilevel programming. In that setting, five classes of generalized convexity for bifunctions are introduced, namely, ∂^* -convexity, strict ∂^* -convexity, ∂^* -pseudoconvexity, strict ∂^* -pseudoconvexity, and ∂^* -quasiconvexity, and complete Wolfe-type duality theories along with weak as well as strong duality theorems are obtained through value-function reformulation.

The dissertation further studies the work of Gadhi and Dempe, in which the implicit lower-level optimality condition is transformed into an explicit scalar inequality through the Ψ -reformulation and the Hiriart-Urruty scalarisation function. This reformulation overcomes the failure of the Mangasarian–Fromovitz constraint qualification that affects the value-function approach and allows KKT-type necessary conditions to be established under a nonsmooth Abadie constraint qualification. Finally, following Saini and Kailey, the Ψ -reformulation and convexifactor calculus are combined to obtain the first complete duality theorems for the bilevel multiobjective problem in the convexifactor framework. The strong duality theorem shows that every primal weak efficient solution satisfying the nonsmooth Abadie constraint qualification gives rise to a weak efficient dual solution with identical objective values.

The findings support the view that convexifactors provide sharper multiplier conditions than the Clarke subdifferential, that the Ψ -reformulation resolves the constraint-qualification difficulties of the value-function approach, and that together these methods provide a natural and consistent framework for nonsmooth hierarchical multiobjective optimisation.

Keywords: bilevel programming, multiobjective optimisation, convexifactors, weak efficiency, Karush–Kuhn–Tucker conditions, Ψ -reformulation, generalized convexity, Wolfe duality, non-smooth Abadie constraint qualification.

Chapter 1

Introduction

1.1 Background & Motivation

Bilevel optimization originates from the study of hierarchical decision-making processes that arise across economics, engineering, transportation, and management science. Such problems model situations in which one decision-maker (the leader) acts while anticipating the optimal reaction of another (the follower), a framework that was mathematically formalized through Stackelberg’s leader-follower theory and subsequently developed into bilevel programming [12, 13]. These problems are structurally distinct from single-level optimization because here the solution set of a parametric lower-level problem determines the feasible region at the upper level; the scalar linear case provided the initial setting for rigorous study.

The natural extension to multiobjective bilevel programming is formed where both leader and follower pursue several competing objectives simultaneously, such as cost, quality, or efficiency. In this setting, the appropriate solution concept shifts from classical minimization to Pareto or weak efficiency, and the feasible regions involved are set-valued and typically nonsmooth, which renders smooth classical analysis insufficient [8, 18]. To handle such problems, tools from nonsmooth analysis are required; in particular, Clarke subdifferentials, Dini directional derivatives, and generalized convexity notions have proved indispensable. Among these, convexifactors—closed sets that encode directional derivative information without requiring convexity—have emerged as especially well-suited instruments for nonsmooth vector optimization [3, 10].

Convexifactors were introduced by Jeyakumar and Luc and further developed by Dutta and Chandra, who showed how they enable both optimality and duality results by providing a framework that generalizes classical monotonicity and pseudoconvexity conditions. The theory was subsequently applied by Suneja and Kohli to scalar bilevel problems and by Gadhi and Dempe to multiobjective bilevel problems through the Ψ -reformulation. The present dissertation draws these threads together, placing them within a unified convexifactor-based single-level reformulation framework that supports the derivation of optimality conditions and duality results.

Motivated by concrete applications in transportation network design, policy modelling, and

engineering systems, and by the observation that classical smooth methods are inadequate when the relevant mappings are set-valued and nonsmooth, this work addresses identifiable gaps in the existing literature by providing an integrated analytical treatment. Convexifactors are placed at the center of the analysis throughout [3, 4, 5].

1.2 Problem Statement

Bilevel multiobjective programming optimization is a hierarchical structure whose upper-level decisions depends on the optimal response of a problem of lower-level. This reliance complicates the problem by itself beyond the case of single-level optimization, especially when managing the nonsmooth objective and constraint functions and both levels are of vector-valued objectives.

The overall objective of the dissertation is to create a convexifactor-based model of multiobjective bilevel optimization that maintains the hierarchical nature of the original problem but allows it to be reformulated, analyzed in terms of optimality, and duality. Specifically, the dissertation aims at achieving a similar single level formulation, establishing multiplier type conditions of weak efficient solutions under appropriate nonsmooth constraint qualification, and building Wolfe type dual problems whose weak and strong duality holds.

Therefore, the main issue under investigation here consists in the creation of a consistent theoretical framework of a nonsmooth multiobjective bilevel optimization based on convexifactors, convexity-generalized variational analysis.

1.3 Aim of the dissertation

The dissertation will focus on establishing a strict theoretical base of multiobjective bilevel optimization with nonsmooth and generalized convexity conditions with convexifactors as the key analytical instrument. It aims to generalize the theory of classical bilevel programming to a more realistic context where the nonsmoothness of hierarchical decision making and the vector nature of the objectives are inherent.

To be more precise, the dissertation explores the possibility of using convexifactor calculus, generalized convexity and nonsmooth constraint qualifications to achieve meaningful reformulations, optimality and duality results, It further seeks to fill the gap between traditional optimization theory and the more realistic models that occur in bilevel and multiobjective optimization, where differentiability, convexity, and regularity can be lost.

An additional goal is to demonstrate that convexifactor-based approaches provide real extensions of the previous scalar and single-objective findings, and not formal ones. Specifically, the dissertation aims to bring together weak efficiency, optimistic bilevel assumptions, lower-level value-function analysis and nonsmooth Abadie-type constraint qualifications into one consistent framework.

Finally, the goal of the work is to consolidate the theory of nonsmooth vector optimization

and hierarchical optimization through the provision of a mathematically consistent approach to reformulation, optimality and duality in bilevel multiobjective programming and a foundation to future work on stability, sensitivity and algorithmic methods.

1.4 Objectives

The dissertation has the following objectives:

1. To come up with a convexifactor-based theoretical framework to study multiobjective bilevel optimization problems with nonsmooth and generalized convexity assumptions.
2. To develop the bilevel multi-objective problem into a single-level problem through lower-level optimality conditions, value-function representations and associated reformulation methods.
3. To examine the structural properties of the lower-level solution mapping, lower-level value, and the reformulated constraints in the convexifactor setup.
4. To obtain optimality conditions needed to find weak efficient solutions of the reformulated problem using convexifactor calculus, generalized convexity and non-smooth constraint qualification.
5. To determine the contribution of the nonsmooth type of Abadie qualification of constraint to meaningful first-order optimality conditions of the reformulated bilevel model.
6. To generalize previous scalar and single-objective bilevel outcomes to the truly multiobjective environment, such that the lower-level and upper-level equations are considered as be vectors as opposed to scalarized formulations.
7. To ensure that the suggested framework is consistent with familiar special cases and simplifies when the problem structure is simpler to classical results.
8. To present an analytical treatment of reformulation, optimality and duality in bilevel multiobjective programming in a unified way that would form a basis on the future work in sensitivity analysis, stability analysis and algorithm development.

1.5 Scope of Study

The study for multiobjective bilevel optimization problems is the theoretical study under the nonsmooth and generalized convexity assumptions of such problems is the scope of this dissertation. The analysis is confined to the construction of a convexifactor-based model that can be applied to reformulate these problems as equivalent at the single-level model and subsequently acquire optimality and duality outcomes based on the reformulated structure. It is not concerned

with numerical experiments, implementation problems or performance of algorithms, but rather with rigorous mathematical foundations and structural findings.

In this framework, the dissertation takes into account hierarchical problems where the objective function are vector-valued for both lower and upper-level such that the meaning of a solution is weak efficiency as opposed to ordinary scalar efficiency. The lower-level solution set, value-function and gap-function representation are considered to be the critical elements of the analysis, as they are the elements that render the bilevel problem nonsmooth and technically challenging. Convexifactors, generalized convexity, and nonsmooth constraint qualifications are thus the primary analytical tools used in the study.

The derivation of the needed optimality conditions of the reformulated problem under nonsmooth Abadie-type constraint qualification is also part of the scope. This is then succeeded by the Wolfe type dual problems construction and the investigation of weak and strong duality, with appropriate assumptions. Thus, the dissertation attempts to bridge the entire chain of theory between the reformulation and the primal optimality and eventually to the duality, yet only in the context of convexifactor-based environment, which is taken in the present paper.

The dissertation is restricted to deterministic bilevel models and not generalized to stochastic, robust and dynamic bilevel models. Neither does it strive to generate computational procedures or benchmark comparisons. This is to develop a clean and coherent analytical theory which they can adhere to in the future to support such developments. This implies that emphasis is laid on equivalence, feasibility, weak efficiency, constraint qualification, and duality instead of approximation or numerical solution schemes.

The other limitation to the study is that the study is meant to extrapolate previous scalar and single-objective bilevel findings to the truly multiobjective environment, where the assumptions supporting the use of convexifactor calculus hold. Consequently, the conclusions are only meant to be applicable to a mathematically significant group of nonsmooth hierarchical problems, but not to all possible bilevel models in the literature. This maintains the scope broad enough to be meaningful, but narrow enough to be manageable.

In general, the dissertation scope will be to develop a coherent theoretical basis of multi-objective bilevel programming using convexifactors with particular emphasis on reformulation, optimality, and duality. The dissertation stays within the boundaries of nonsmooth deterministic hierarchical optimization and aims to provide a foundation for future work in sensitivity analysis, stability analysis, and algorithm design.

1.6 Limitations

This dissertation is a coherent convexifactor-based model of multiobjective bilevel optimization, but the scope of the study is inherently restricted due to the theoretical nature of the study and the assumptions made to derive the results. It works on the topics of reformulation, optimality, and duality in a deterministic nonsmooth context, and thus does not seek to encompass all the

possible variants of bilevel optimization that occur in the larger literature. Specifically, the dissertation is devised to deal with the types of problems where the convexfactor calculus, the generalized convexity, and the nonsmooth constraint qualifications make sense and are technically implementable.

A weakness of the dissertation is that it does not build numerical algorithms or computational experiments. Consequently, it lacks convergence tests, implementation details, and benchmark comparisons to solving the reformulated problems in practice. It is all about analytical results, and thus the dissertation sets the theoretical base of future computational work and not a full-fledged solver-based methodology. This implies that the questions of efficiency, scalability, and numerical performance are out of bounds as yet.

The second weakness is that the findings require the presence of appropriate convexfactor representations and generalized convexity assumptions. These assumptions are strong, but not applicable to all nonsmooth functions, nor to all hierarchical models. This means that the dissertation findings cannot be generalized to the solutions of arbitrary multiobjective bilevel problems automatically, in particular in cases when the solution mapping of the lower-level is highly irregular, discontinuous, or nonconvex in a manner that is not within the framework adopted.

The dissertation is primarily a deterministic setting, though. It is not concerned with stochastic uncertainty, robust bilevel models, or dynamical decision structures, or with bilevel models based on learning on a large scale. These issues may need varying tools of analysis and have other complications like randomness, adversarial behavior and time-dependence. Due to this reason, the current theory is to be considered as a fundamental input to a reduced deterministic model and not a general theory of all the bilevel models.

The other weakness is that the reformulation strategy can add auxiliary conditions and assumptions that are mathematically convenient but not necessarily easy to check in practice. In particular, the nonsmooth Abadie-type constraint qualification and the lower-level optimality representation are the focus of the analysis, and the verification of these conditions in specific models can involve more effort. Similarly, the results of duality are obtained by making assumptions that guarantee a mathematical validity, and these assumptions can be used to limit the set of problems to which the theorems apply directly.

Lastly, the dissertation generalizes previous scalar and single-objective findings to a truly multiobjective bilevel model, but again this is a theoretical generalization and not a fully developed practical decision-making model. The work explains the problem structure and gives the results on equivalence, optimality and duality, but the translation of the results to the algorithmic procedures, software implementation, or application-specific model is a subject of further research. In this, the dissertation ought to be perceived as a foundational study opening a few directions upon further research as opposed to a definitive treatment of the topic.

1.7 Research Questions

Research Questions

The following broad research questions guide this dissertation:

1. In what ways is it possible to construct a convexfactor-based framework to multiobjective bilevel optimization problems in such a way that the key properties of optimization can be maintained even when nonsmoothness and hierarchical dependence is present?
2. How can the original multiobjective bilevel equations be expressed as an single-level equation in terms of the lower-level solution mapping, or a gap-function representation?
3. What does convexfactors, convexity as a generalized notion and nonsmooth constraint qualifications provide us with meaningful conditions of necessity and optimality for weak efficient solutions of the reformulated equation?
4. What are the constructions of Wolfe-type dual problems of the reformulated bilevel model and what are the conditions under which weak and strong duality results are true?
5. How far can the proposed results be compared with known scalar or smooth bilevel optimization results, and how can they be used to extend the work on sensitivity analysis, stability analysis, and the development of algorithms?

1.8 Dissertation Outline

This dissertation is structured into seven chapters. Chapter 1 introduces the background, motivation, problem statement, aims, objectives, scope, limitations, and research questions. Chapter 2 entails the preliminaries and the methodology that will be applied throughout the dissertation. Chapter 3 is a literature review of convexfactors, nonsmooth optimization, bilevel programming and duality. Chapter 4 elaborates the theory of convexfactors, generalized convexity and vector maximization. In chapter 5, optimality and duality of a scalar bilevel programming with convexfactors are studied. It also derives the necessary optimality conditions via a (Psi)-reformulation. Chapter 6 provides duality results to the bilevel multiobjective problem in case of convexfactors. Lastly, Chapter 7 offers a comparative analysis of the four main papers and ends the dissertation by discussing its main contributions and scope.

Chapter 2

Preliminaries and Methodology

2.1 Preliminaries

Preliminary 2.1. Locally Lipschitz Function [1] A function p from $\mathbb{R}^n$ to $\mathbb{R}$ is called locally Lipschitz at a if for $a \in \text{neighbourhood } U$ and some constant $K \geq 0$ such that

$$|p(b) - p(c)| \leq K \|b - c\| \quad \forall b, c \in U.$$

Preliminary 2.2. Upper Dini Directional Derivatives and Lower Directional Derivatives [2]

Let $p : \mathbb{R}^n \rightarrow \mathbb{R}$ be finite at a . In the direction of v , the lower Dini derivative of p at a is

$$p'_-(a; v) = \liminf_{t \downarrow 0} \frac{p(a + tv) - p(a)}{t},$$

and the upper Dini derivative is

$$p'_+(a; v) = \limsup_{t \downarrow 0} \frac{p(a + tv) - p(a)}{t}.$$

Preliminary 2.3. Clarke Generalized Subdifferential and Jacobian [1] Let p be a function from $\mathbb{R}^n$ to $\mathbb{R}$ be locally Lipschitz at a . Then for p at a , Clarke Generalized subdifferential is defined as the convex hull for all limit points of gradients $\nabla p(a_k)$ which is denoted by $\partial^C p(a)$, where $a_k \rightarrow a$ and p is differentiable at each a_k .

For a locally Lipschitz mapping $P : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Clarke generalized Jacobian $\partial^C P(a)$ is derived from the convex hull of all limit points of Jacobian matrices $JP(a_k)$, where $a_k \rightarrow a$ and P is differentiable at each a_k .

Preliminary 2.4. Chain Rules for Convexifactors [3] Let $P : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be continuously differentiable and let q be a function $\mathbb{R}^m \rightarrow \mathbb{R}$ be a continuous function.

If q admits a bounded convexifactor at each point, then $q \circ P$ admits an USRCF at a given by

$$\partial(q \circ P)(a) = J_P(a)^T \partial q(P(a)).$$

If q does not admit bounded convexifactors, then for any $\varepsilon > 0$, $q \circ P$ admits an upper

semi-regular convex factor at m of the form

$$J_P(m)^T B \in \partial q(P(m)),$$

where B denotes the unit ball in the appropriate space.

Preliminary 2.5. Bilevel Multiobjective Programming Problem [4, 5] Consider the problem

$$(P) \quad \min_{a,b} F(a, b) = (F_1(a, b), F_2(a, b), \dots, F_n(a, b))$$

subject to

$$G_l(a, b) \leq 0, \quad l \in L, \quad b \in S(a),$$

where $S(a)$ at the lower-level problem is the solution set.

Preliminary 2.6. Lower-Level Problem (P_a) [4, 5] For each fixed a , the lower-level problem is

$$(P_a) \quad \min_b q(a, b)$$

subject to

$$g_s(a, b) \leq 0, \quad s \in S.$$

Preliminary 2.7. Feasible Set $Y(a)$ or $S(a)$ [4, 5] For each fixed $a \in \mathbb{R}^{n_1}$, a feasible set is known as:

$$Y(a) = \{ b \in \mathbb{R}^{n_2} : g_s(a, b) \leq 0, s \in S \}.$$

The solution set of the lower-level problem is

$$S(a) = \arg \min \{ q(a, b) : b \in Y(a) \}.$$

Preliminary 2.8. Optimal Value Function $V(a)$ [4, 5] It is defined by

$$V(a) = \{ q(a, b) : b \in Y(a) \}.$$

for the lower-level problem

Preliminary 2.9. Weak Efficient Solution [8, 4] A feasible point (a, b) of (P) is defined as weak efficient solution if there does not exist another feasible point $(\bar{a}, \bar{b})$ such that

$$F_i(\bar{a}, \bar{b}) < F_i(a, b) \quad \text{for all } i = 1, 2, \dots, n.$$

If this condition holds only in a neighborhood, then (a, b) is known as the local weak efficient solution.

Preliminary 2.10. Optimistic Bilevel Assumption [4, 5] In the optimistic formulation, it is assumed that for a fixed leader's decision a , the follower selects a solution $b \in S(a)$ that is most favorable for the upper-level objective.

Preliminary 2.11. Gap Function $\psi(a, b)$ or Ψ -Function [4, 5] For each fixed a , define

$$\psi(a, b) = \max_{z \in Y(a)} \min \left\{ t(a, b) - t(a, z), \max_{s \in S} t_s(a, z) \right\}.$$

This function converts optimality of decision at the lower level into a single scalar inequality.

Preliminary 2.12. Hiriart-Urruty Gauge $C(A, \mathbb{R}_+^q)$ [6] Let $C \subset \mathbb{R}^q$ be a closed, convex cone with nonempty interior. The gauge associated with C is the function

$$C(y) = d(y, \mathbb{R}^q \setminus C),$$

where $d(\cdot, \cdot)$ denotes the distance function. This function is convex, positively homogeneous, and 1-Lipschitzian.

2.2 Notations

The following notation is adopted throughout the entire dissertation unless explicitly stated otherwise.

The symbol $\mathbb{R}^n$ denotes the n -dimensional Euclidean space equipped with the standard notation for the inner product $\langle \cdot, \cdot \rangle$ and the induced norm $\| \cdot \|$. For a point $x \in \mathbb{R}^n$ and radius $\varepsilon > 0$, the open ball centred at x with radius ε is denoted $B(x, \varepsilon)$, and the closed unit ball in $\mathbb{R}^n$ is denoted $B_{\mathbb{R}^n}$.

For a set $A \subseteq \mathbb{R}^n$, its closure, convex hull, and closed convex hull are written $\text{cl } A$, $\text{co } A$, and $\overline{\text{co}} A$, respectively; its negative polar cone is A° . If $S : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ is a set-valued map, $(\text{co } S)(x) := \text{co}(S(x))$.

The space of all linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ is denoted $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ with closed unit ball $B_{\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)}$. If $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at x , its Jacobian matrix is written $JF(x)$.

For the bilevel problems studied in Chapters 4–7, the upper-level decision variable is $a \in \mathbb{R}^{n_1}$ (leader) and the lower-level decision variable is $b \in \mathbb{R}^{n_2}$ (follower), with $n_1, n_2 \geq 1$. Upper-level objective functions are denoted F_i or p , lower-level objectives by p or ϕ , ℓ_s , or γ_i , with index sets I, J, S as defined in each chapter. The optimal value function of the lower-level problem is $V(a) := \inf\{p(a, b) : b \in Y(a)\}$, where $Y(a)$ denotes the lower-level feasible set for fixed a , and $S(a)$ (or $\Lambda(a)$, $O(a)$) denotes the corresponding solution set.

Convexifactors are denoted $\partial^* F$ or ∂F (with superscripts u, l, us for upper, lower, and upper semi-regular variants as in Chapter 3). Clarke subdifferentials are written $\partial^C f$ or $\partial^\circ f$. The contingent (Bouligand tangent) cone to a set E at a point $\bar{u}$ is $K(E, \bar{u})$, and the normal cone is

$N(E, \bar{u})$. The Hiriart-Urruty signed-distance scalarisation function for a set S is Δ_S , as recalled in Preliminary 5.2.

All other notation is introduced locally where it first appears.

2.3 Methodology

The dissertation follows a theoretical and analytical methodology.

This dissertation adopts a theoretical and analytical approach to examine multi-objective bilevel maximization problems under nonsmoothness and generalized convexity. It is not based on numerical experiments or simulations, but instead develops a rigorous mathematical framework using convexifactors, reformulation techniques, and generalized convexity. The key approach is to reformulate the bilevel problem into an equivalent single-level problem and then derive optimality and duality results for the reformulated model.

The first step is to establish the theoretical foundation required in the subsequent chapters. These include locally Lipschitz functions, directional derivatives, generalized subdifferentials, convexifactors, generalized convexity, weak efficiency, and cone-based constraint qualifications. These concepts form the mathematical basis of the dissertation, particularly the concept of convexifactors, which provides a convenient set-valued framework for describing nonsmooth behavior in bilevel programming.

The second step is the reformulation of the initial bilevel problem. It is not directly tractable since the feasible set depends on the lower-level solution set for the upper level. To address this, the dissertation employs value function or gap-type reformulations from the literature, reducing the bilevel problem to a single-level problem with additional nonsmooth constraints.

The third step is the application of convexifactor calculus. The composite functions that arise from the bilevel structure are handled using chain rules, mean-value results, and support-function arguments. This is necessary because the objectives, value functions, and reformulated constraints may become nonsmooth, even when the original data are smooth.

The dissertation then derives necessary optimality conditions for weak efficient solutions. These are established under generalized convexity assumptions and a suitable nonsmooth Abadie-type constraint qualification. Under these conditions, multiplier rules analogous to classical KKT theory can be derived for the nonsmooth vector bilevel setting.

The methodology also develops Wolfe-type duality results for the reformulated problem. Under the same generalized convexity and constraint qualification framework, weak and strong duality can be established. This provides structural insight into the primal model and ensures consistency between the primal and dual formulations.

Finally, the methodology verifies consistency with known scalar and simpler special cases and follows a logical chapter-wise progression. Chapter 2 develops convexifactor tools, Chapter 3 investigates reformulation, Chapter 4 derives optimality conditions, and Chapter 5 develops duality results. Thus, the dissertation is organized in a systematic sequence of preliminaries,

reformulation, optimality, and duality.

Chapter 3

Literature Review

3.1 Introduction

The theory of optimality & duality in bilevel multiobjective programming under nonsmooth and generalised convexity assumptions draws on several research strands in mathematical optimisation that have largely developed in parallel over the past five decades. This chapter reviews the relevant literature in a concept-driven manner, tracing the development of each central idea—convexifactors, optimality and duality in multiobjective programming, conditions for nonsmooth problems, bilevel reformulation methods, and duality in bilevel multiobjective programming—and situating the contributions of the four principal papers of this dissertation within the broader research context. Rather than surveying one paper at a time, the review deliberately weaves together complementary lines of inquiry, emphasising how advances in one area have enabled progress in another, and how the four papers form a coherent theoretical programme.

3.2 Convexifactors in Nonsmooth Optimisation

Classical optimisation theory for years assumed that objective and constraint functions are differentiable and continuous, which makes it possible to characterise stationary points via gradients and to write down multiplier rules. Many practical models, however, involve functions that do not satisfy this condition: the optimal value of lower-level programme, the pointwise maximum over a parametric family, and piecewise-smooth data are all familiar sources of nonsmoothness. The development of subdifferential calculus was motivated precisely by the need to extend optimality theory beyond the smooth case.

The Clarke generalised subdifferential [1] is the most widely employed such object. For a locally Lipschitz function p from $\mathbb{R}^n \rightarrow \mathbb{R}$, $\partial^C f(\bar{x})$ being constructed as the convex hull of all accumulation points of gradients $\nabla f(x_k)$ along sequences $x_k \rightarrow \bar{x}$ at which f is differentiable. The associated Clarke directional derivative $f^\circ(\bar{x}; v)$ furnishes an upper bound on all one-sided rates of change, and the attendant calculus rules for sums, chains, and maxima make the theory computationally tractable [1, 2]. However, the compactness and convexity of $\partial^C f(\bar{x})$ can be a

drawback: for certain nonsmooth functions the Clarke subdifferential is strictly larger than any set that adequately captures directional behaviour, so optimality conditions derived from it may be unnecessarily weak.

The desire for sharper, and possibly non-convex, derivative-like objects prompted the investigation of convexifactors. The Michel–Penot subdifferential [20], defined through a different directional derivative, is one early instance. A systematic framework was then established by Jeyakumar and Luc [19], who introduced a closed set $\partial p(\bar{x})$ as *convexifactor* of p at $\bar{x}$ when the upper Dini directional derivative satisfies

$$p_d^+(\bar{x}; v) = \sup_{\xi \in \partial p(\bar{x})} \langle \xi, v \rangle \quad \forall v \in \mathbb{R}^n,$$

together with corresponding upper convexifactors, lower convexifactors, and upper semi-regular convexifactors defined through inequalities involving the upper and lower Dini derivatives. Crucially, a convexifactor is not required to be convex or compact, so it can be genuinely smaller—and therefore yield sharper conditions—than the Clarke subdifferential. Jeyakumar and Luc also established a mean-value theorem for convexifactors that plays the same structural role as the classical mean-value theorem in smooth analysis: it bridges local derivative information at individual points with global function differences and underpins both chain rules and optimality conditions. The existence of compact convexifactors under weak boundedness conditions was studied by Uderzo [21], who related convexifactors to the earlier theory of convex approximators and exhausters.

The convexifactor calculus was completed as an instrument for optimisation in the first of the four main papers of this dissertation, by Dutta and Chandra [3]. They established two chain rules: one for the case in which the outer function admits a bounded convexifactor (yielding an upper semi-regular convexifactor for the composition), and one for the unbounded case (yielding an ε -perturbed upper semi-regular convexifactor). These chain rules are indispensable when a bilevel problem is reduced to a single-level formulation, since the resulting constraints are typically compositions of nonsmooth functions. Dutta and Chandra also clarified the strict hierarchy

$$\text{upper regular CF} \Rightarrow \text{upper semi-regular CF} \Rightarrow \text{upper CF},$$

providing counterexamples to show that no reverse implication holds.

The relationship between convexifactors and the Clarke subdifferential is central to understanding the scope of the theory. As noted in [3, 19], the Clarke subdifferential of any locally Lipschitz function is itself a convexifactor; so too is the Michel–Penot subdifferential [20]. The convex hull of CF may, however, being strictly smaller than the Clarke subdifferential, and this is precisely the mechanism through which convexifactors can produce sharper multiplier conditions. The early framework was extended further by Demyanov [44] and by Demyanov and Jeyakumar [45] in their study of convexificators (a related but distinct concept), who explored

the question of finding the smallest convex subdifferential-like object for a given function.

Taken together, these contributions establish convexifactors as a mathematically mature refinement of classical nonsmooth subdifferential theory, providing the language in which the optimality and duality results developed later in this dissertation are expressed.

3.3 Optimality Conditions in Nonsmooth Optimisation

Once a rigorous derivative-like object is available for nonsmooth functions, the next step is to use it to derive necessary and sufficient conditions for optimality. The classical KKT conditions, which assert the existence of Lagrange multipliers satisfying a stationarity inclusion and complementary slackness, can be extended to the nonsmooth setting by replacing gradients with appropriate subdifferentials.

For scalar problems, Clarke himself derived optimality conditions based on his subdifferential [1], and Hiriart-Urruty [6] subsequently introduced the signed-distance (gauge) function Δ_S , which reduces vector optimality conditions to scalar ones—a device that is fundamental to the Gadhi–Dempe framework for bilevel multiobjective programming [4]. For inequality-constrained problems, KKT-type conditions additionally require a constraint qualification to guarantee that the multiplier being nonzero when associated with the objective. The ACQ, which identifies the contingent (Bouligand tangent) cone with the linearisation cone, is the most minimal such condition in the smooth setting; its nonsmooth extension, in which gradients are replaced by convexifactors or subdifferentials, constitutes the nonsmooth Abadie constraint qualification (NACQ) employed in [4] and [5].

The paper by Dutta and Chandra [3] established necessary optimality conditions for vector minimisation problems within the convexifactor framework. Their principal results (Theorems 3.8–3.10 of Chapter 3 of this dissertation) assert that at any weak efficient point, nonnegative multipliers λ_j , not all zero, exist such that

$$0 \in \sum_{j=1}^k \lambda_j \operatorname{co} \partial^{us} p_j(\bar{a}) + N(C, \bar{a}),$$

where $\partial^{us} p_j(\bar{a})$ denotes the upper semi-regular convexifactor of the j -th objective and $N(C, \bar{a})$ is the normal cone to the feasible set. A notable structural flexibility is that upper semi-regularity may be relaxed for at most one objective component without invalidating the result, a feature absent from the Clarke subdifferential approach. Wang and Jeyakumar [24] established sharp Lagrangian multiplier rules for nonsmooth inequality-constrained problems, demonstrating that convexifactors can produce strictly sharper conditions than the Clarke theory. Luc [23] obtained multiplier rules for multiobjective problems with continuous data using an alternative scalarisation, while Pappalardo and Stocklin [22] studied necessary conditions in nondifferentiable vector optimisation from a topological perspective.

For bilevel problems, necessary optimality conditions are considerably harder to obtain because of the complicated feasible set. Early contributions by Ye and Zhu [32, 35], Dempe [29, 30], and Yezza [33] established first-order conditions for scalar bilevel problems via value function representations. As noted by Gadhi and Dempe [4], however, the value function constraint $\phi(a, b) - V(a) \leq 0$ has a fundamental drawback: at every feasible point the Mangasarian–Fromovitz constraint qualification fails, necessitating the separate hypothesis of *partial calmness* introduced by Ye and Zhu. The Ψ -reformulation approach of Gadhi and Dempe [4] circumvents this difficulty by encoding the lower-level optimality condition through a scalarised auxiliary function.

Babahadda and Gadhi [28] applied convexifactors to obtain KKT-type necessary conditions for optimisation of scalar problems, establishing a link between the Clarke-based multiplier theory and convexifactor calculus. This programme was extended to the bilevel single-objective setting by Suneja and Kohli [?], who also established sufficient conditions and convexity notions within the convexifactor framework. The sufficient conditions in [7] are expressed in terms of ∂^* -convexity, ∂^* -pseudoconvexity, and ∂^* -quasiconvexity (recalled in Chapter 4 of this dissertation), and they do not require any constraint qualification, thus complementing the necessary conditions which do.

The extension of sufficient optimality conditions to the multiobjective bilevel setting was undertaken by Gadhi, Hamdaoui and El Idrissi [43], and, within the Ψ -reformulation and convexifactor framework, by Saini and Kailey [5], the most recent of the four main papers. Chuong [41] obtained optimality conditions for nonsmooth multiobjective bilevel problems using limiting subdifferentials and the extremal principle, providing a point of comparison with the convexifactor approach. The advantage of convexifactors is that they need not be convex and can be smaller than the Clarke or limiting subdifferentials in specific instances, potentially yielding tighter optimality conditions, while their calculus (mean-value theorem, chain rules) remains tractable.

3.4 Duality in Multiobjective Programming

Duality theory provides, for any given optimisation problem (known as the primal), an associated problem (known as the dual) whose objective value furnishes a bound on the primal. In the scalar setting the Wolfe and Mond–Weir dual are most common formulations, differing in the structure of the dual objective and in the strength of the convexity conditions required. Extensions to the multiobjective setting are nontrivial because classical optimality must be replaced by *weak efficiency* (no feasible point strictly improves all objective components simultaneously) or by stronger notions such as Pareto efficiency and proper efficiency [8].

Scalarisation, in which a weighted sum or other scalar functional converts the vector problem into a family of scalar problems, is one route to multiobjective duality. Hiriart-Urruty [6] introduced the signed-distance function Δ_S as a device for scalarising cone-ordered vector

problems; this function is convex, positively homogeneous, and 1-Lipschitz, and it satisfies $0 \notin \partial^{ca} \Delta_S(y)$ for all y —a non-vanishing property that ensures the objective multiplier is nonzero in the resulting optimality conditions. Analogous techniques were employed by Amahroq and Taa [36] and Ciligot-Travain [38] to derive Lagrange–KKT conditions for multiobjective problems, providing the theoretical foundation on which the Gadhi–Dempe paper builds.

Duality in multiobjective programming under smooth convexity assumptions is classical; the standard reference is Bazaraa, Sherali and Shetty [11]. Extensions to nonsmooth functions via Clarke subdifferentials were developed during the 1990s and 2000s, with Amahroq and Gadhi [10] obtaining Lagrange multiplier rules for nonsmooth vector optimisation problems.

The incorporation of convexifactors into multiobjective duality was carried out by Dutta and Chandra [3], who derived multiplier conditions for weak efficient points of vector minimisation problems and showed that the resulting conditions can be sharper than those obtained via the Clarke subdifferential. Li and Zhang [9] investigated optimality conditions for nonsmooth problems under generalised convexity from a complementary viewpoint, while Luc [23] derived multiplier rules through a different convex scalarisation.

The first systematic Wolfe-type and Mond–Weir-type duality treatment of bilevel single-objective programming using convexifactors was given by Suneja and Kohli [7], who established weak and strong duality under ∂^* -convexity and ∂^* -pseudoconvexity, respectively. Their work demonstrated that the convexifactor machinery is sufficiently rich to support a complete duality theory in the bilevel case. The most comprehensive treatment of duality in the multiobjective (non-bilevel) setting via convexifactors prior to the papers in this dissertation is in the work of Dutta and Chandra [34], which provides optimality conditions and the technical scaffolding on which the bilevel extensions are built.

One of the observations that is of special interest to the duality results of Saini and Kailey [5] is the interrelation among the generalised convexity classes: ∂^* -convexity implies both ∂^* -pseudoconvexity and ∂^* -quasiconvexity, but the latter two are independent. The Wolfe-type dual requires ∂^* -convexity of all data functions at the dual point, whereas the Mond–Weir dual (in the scalar bilevel case of [7]) requires only ∂^* -pseudoconvexity of the objective and ∂^* -quasiconvexity of the constraints, allowing it to be applied to weaker assumptions. This hierarchy of assumptions is a common motif in the literature on nonsmooth. multiobjective duality.

3.5 Bilevel Programming and Reformulation Techniques

Bilevel programming developed from the economic theory of hierarchical decision-making. Von Stackelberg [12] described a leader-follower market model in which one party (the leader) makes decisions according to the response of another (the follower). The mathematical formalisation of this idea as a bilevel programme, was developed by Bracken and McGill and subsequently studied by Bard [14, 15] and Vicente [17]. The survey by Colson, Marcotte, and Savard [16]

provides a comprehensive account of the state of the field as of the mid-2000s.

The principal analytical difficulty in bilevel programming is the implicitness of the feasible region, which depends on the upper-level decision in a potentially discontinuous and non-convex manner. Two canonical reformulation strategies have been developed to address this implicitness.

The first is the *value function reformulation*, whereby the optimality constraint $b \in S(a)$ at the lower-level (where $S(a)$ denotes the lower-level solution set for fixed a) is replaced by the scalar inequality. This method was employed by Ye and Zhu [32], Dempe [29, 30], and Ye [46]. Although mathematically transparent, the value function reformulation has a well-known deficiency: the constraint qualification needed for KKT-type conditions (specifically, the Mangasarian–Fromovitz condition) never holds at feasible points, because the constraint $\phi(a, b) - V(a) \leq 0$ is active everywhere and its gradient in the a -direction involves the subdifferential of V , which typically points inward [31].

The second strategy is the Ψ -*reformulation*, introduced by Gadhi and Dempe [4] and subsequently adopted by Lafhim *et al.* [42] and Gadhi, Hamdaoui, and El Idrissi [43]. For a compact set Θ containing the lower-level feasible set $Y(a)$ in a neighbourhood of the point of interest, one defines

$$\psi(a, b, c) := \min \left\{ p(a, b) - p(a, c), -\Delta_{(-\mathbb{R}_+^\ell)}(q_1(a, c), \dots, q_\ell(a, c)) \right\},$$

and the Ψ -function $\Psi(a, b) := \max_{c \in \Theta} \psi(a, b, c)$. The Hiriart-Urruty scalarisation function $\Delta_{(-\mathbb{R}_+^\ell)}$ encodes the feasibility of c for the lower-level problem. Key lemmas in Chapter 5 of this dissertation, following [4], establish that $\Psi(a, b) \leq 0$ for all feasible (a, b) , with equality if and only if $b \in S(a)$. Crucially, the Mangasarian–Fromovitz condition can hold at feasible points of the Ψ -reformulated problem, enabling KKT-type necessary conditions to be derived without requiring partial calmness [4].

In the optimistic formulation of bilevel programming, the follower is assumed to select, from among the optimal lower-level responses, the one most favourable to the leader. This is the standard framework within which the reformulation strategies above are implemented [31, 14]. Under this assumption, the bilevel equation and its single-level reformulation both being locally equivalent at any weak efficient solution, which rigorously justifies analysing the reformulated problem in place of the original.

Bilevel programming finds applications in transportation network design [18], environmental economics, principal-agent problems in contract theory, and machine learning, including hyperparameter optimisation and meta-learning. Recent surveys document the expanding reach of bilevel methodology into data-science applications and provide comprehensive bibliographies of current research.

3.6 Duality in Bilevel Multiobjective Programming

The combination of multiple objectives with bilevel structure is the most demanding setting considered in this dissertation, and the literature on it is correspondingly more recent and less complete than that on either topic in isolation. Necessary optimality conditions for multiobjective bilevel problems were obtained by Gadhi and Dempe [4] using the Ψ -reformulation and Clarke subdifferentials; the corresponding sufficient conditions and duality results were then established by Gadhi, Hamdaoui, and El Idrissi [43] under additional convexity assumptions within the same Ψ -reformulation framework.

The first complete duality treatment for bilevel programmes (in the scalar case) using convexifactors was given by Suneja and Kohli [7], who formulated Wolfe-type dual as well as Mond-Weir-type duals with complete weak and strong duality theorems. This represented a significant advance over earlier work on bilevel duality [37, 39], which had been confined to smooth or Clarke-subdifferential frameworks.

The culmination of this line of development is the paper of Saini and Kailey [5], which establishes both duality theorems, weak and strong, for the bilevel multiobjective problem in the convexifactor framework. It combines the Ψ -reformulation of [4] with the convexifactor calculus of [3, 19] and the generalised convexity framework of [7]. Specifically, Saini and Kailey define a Wolfe-type dual and prove that (i) weak duality holds under ∂^* -convexity of all data functions at any dual-feasible point, and (ii) strong duality holds at any primal weak efficient solution at which the NACQ holds, yielding a dual feasible point with equal primal and dual objective values that is itself a weak efficient solution of the dual.

Lafhim *et al.* [42] had earlier obtained necessary optimality conditions for these problem through the Ψ -reformulation without convexifactors; Gadhi *et al.* [43] then added sufficient conditions and a partial duality theory. Saini and Kailey [5] can accordingly be regarded as completing that programme by incorporating convexifactors into the duality analysis, which potentially sharpens the conditions compared with those available from the Clarke subdifferential. A complementary construction using limiting subdifferentials and the extremal principle was carried out by Chuong [41] for nonsmooth multiobjective bilevel problems in infinite-dimensional spaces; the convexifactor method of [5] operates in finite dimensions and may yield tighter conditions there.

Bao, Gupta, and Mordukhovich [37] obtained necessary conditions for multiobjective optimisation with equilibrium constraints using advanced variational analysis tools including the Mordukhovich subdifferential and the extremal principle. Their work offers a significant point of comparison: while the Mordukhovich subdifferential is the smallest closed subdifferential-like object, computing it can be demanding, whereas convexifactors are more tractable in certain problem classes.

3.7 Summary of Important Developments

The above exposition shows a logical and linear flow of thought leading to the main subject of this dissertation.

1. **Generalized nonsmooth calculus.** The Clarke generalized subdifferential [1] and the Dini directional derivatives are the initial rigorous tools for handling nonsmooth optimization functions. The Hiriart-Urruty scalarization function [6] made it possible to convert vector optimality conditions into scalar ones.
2. **Introduction of convexifactors.** Jeyakumar and Luc [19] introduced convexifactors as adaptable, possibly non-convex generalizations of subdifferentials, and demonstrated calculus based on the mean-value theorem. Dutta and Chandra [3] developed the calculus further, including chain rules and hierarchy among the different types of convexifactors, and used to obtain the conditions necessary for vector optimization.
3. **Reformulation and necessary conditions for bilevel programming.** Bracken and McGill [13], Bard [14, 15], and Ye and Zhu [32] laid the foundation for the value-function reformulation and its related optimality conditions for scalar bilevel problems. The Ψ -reformulation was proposed by Gadhi and Dempe [4] for multiobjective bilevel problems, in order to bypass the failure of the Mangasarian–Fromovitz condition that affects the value-function method.
4. **Duality in bilevel programming by convexifactors.** The Wolfe-type and Mond–Weir-type duals were developed by Suneja and Kohli [7] for scalar bilevel programming using convexifactors. Gadhi et al. [43] established sufficient conditions and partial duality results for the multiobjective case of the Ψ -reformulation.
5. **Culmination: multiobjective bilevel duality through convexifactors.** All three threads, namely Ψ -reformulation, convexifactor calculus, and multiobjective duality, were combined by Saini and Kailey [5], resulting in the first complete duality theorems both weak and strong for the bilevel multiobjective problem in the convexifactor setting. The present dissertation elaborates on, contextualizes, and builds on this body of work.

These five steps are directly related to the five chapters of analysis in this dissertation (Chapters 3–7), establishing the fact that the four papers selected jointly form a self-consistent theoretical program. The gaps identified in the review—the lack of numerical algorithms, the restriction to deterministic finite-dimensional settings, and the reliance on the optimistic bilevel assumption—are recognized as guidelines for future research in Chapter 7.

Chapter 4

Convexifactors, Generalized Convexity and Vector Optimization

4.1 Introduction

A key instrument of nonsmooth analysis and generalized convexity, the theory of convexifactors is the main analytical tool of this dissertation. Although convexity can play a significant role in classical optimization by simplifying analysis and supporting duality and optimality results, many practical problems involve nonsmooth functions and hierarchical structures where classical smooth methods are inadequate. These ideas are extended to provide a structured method for analyzing nonsmooth functions using directional information [3].

The need for convexifactors is explained by the fact that most optimization functions, such as pointwise maxima and minima, absolute value functions, piecewise-defined functions, and lower-level value functions, are differentiable only at a few points, but still have meaningful directional behaviour. Convexifactors provide a flexible set-valued model at such points, capturing this behaviour and allowing further analysis. Dutta and Chandra [3] worked out a unified framework comprising nonsmooth functions and generalized convexity through convexifactors, along with basic calculus and applications to vector optimization.

A significant benefit of this framework is that it transports important concepts of smooth analysis to the nonsmooth setting. Convexifactors, in lieu of gradients, enable the application of chain rules, mean-value arguments, and generalized optimality conditions without the need for differentiability. This renders them especially useful for composite functions and lower-level value functions that naturally arise in bilevel programming [3, 1, 2]. Their relation to generalized convexity is also important, as notions such as pseudoconvexity and quasiconvexity can be studied within the same framework and help in identifying situations where local optimality implies global optimality [3, 9, 10].

The framework also works in the context of vector optimization, where optimality is defined in terms of weak efficiency or Pareto-type concepts as opposed to a single scalar minimum. Dutta and Chandra provided conditions for weak efficient solutions and obtained multiplier-type results in the vector setting, demonstrating that convexifactors provide a rigorous basis for multiobjective analysis. They also facilitate the treatment of nonsmooth constraints and feasible-set geometry using closed-set and support-function descriptions, giving optimality

conditions that are similar to classical results.

The chapter gives the theoretical background for the subsequent analysis of bilevel multiobjective optimization. Although the underlying data may be smooth, bilevel models tend to produce nonsmooth objects such as value functions, reaction mappings, and reformulated constraints. This is the reason why the chapter introduces upper and lower Dini derivatives, upper and lower convexifactors, and upper semi-regular convexifactors, along with their role in generalized convexity and vector optimization based on a specific criterion of interest. Such concepts are applied in the rest of the dissertation since they provide the greater flexibility required in comparison with classical subdifferentials and are especially well suited to the reformulation and analysis of optimality and duality developed in the subsequent chapters of this dissertation [4, 7, 5, 11, 3].

4.2 Preliminaries

All notation used in this chapter follows the unified conventions established in Section 2.2 of Chapter 2. In particular, $\mathbb{R}^n$ denotes the standard n -dimensional Euclidean space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$; $\text{co } A$, $\overline{\text{co } A}$, $B_{\mathbb{R}^n}$, $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, and $JF(x)$ retain their meanings from that section. The foundational analytic concepts recalled below are required for the results of this chapter.

Preliminary 4.1. Dini Directional Derivatives[3]

Let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$ be finite at $a \in \mathbb{R}^n$. The *upper Dini directional derivative* of p at a in the direction $v \in \mathbb{R}^n$ is

$$p^\uparrow(a; v) = \limsup_{t \rightarrow 0^+} \frac{p(a + tv) - p(a)}{t},$$

& the *lower Dini directional derivative* for the same is defined by

$$p^\downarrow(a; v) = \liminf_{t \rightarrow 0^+} \frac{p(a + tv) - p(a)}{t}.$$

In general, these quantities may take values in $\mathbb{R} \cup \{-\infty, +\infty\}$. If p is locally Lipschitz at a , then both derivatives are finite for every $v \in \mathbb{R}^n$. Moreover,

$$p^\downarrow(a; v) \leq p^\uparrow(a; v)$$

for all $v \in \mathbb{R}^n$, and equality for every v holds precisely when the ordinary directional derivative $p'(a; v)$ exists.

Preliminary 4.2. Clarke Generalized Subdifferential [11] Let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$

be locally Lipschitz at a . *Clarke generalized subdifferential* of p at a is

$$\partial^C p(a) = \text{co} \left\{ \lim_{k \rightarrow \infty} \nabla p(a_k) \mid a_k \rightarrow a, p \text{ is differentiable at each } a_k \right\}.$$

It is known that $\partial^C p(a)$ is convex and compact, and that the Clarke directional derivative

$$p^\circ(a; v) = \max_{\xi \in \partial^C p(a)} \langle \xi, v \rangle$$

satisfies

$$p^\downarrow(a; v) \leq p^\circ(a; v)$$

for all $v \in \mathbb{R}^n$.

4.3 Definitions and Calculus Rules of Convexifactors [3]

4.3.1 Hierarchy of Convexifactors

We now introduce the central objects of this chapter. The following definitions follow the framework of Jeyakumar and Luc [19] and the development of Dutta and Chandra [3].

Definition 4.1. Upper Convexifactor [19, 3]

Let $p: \mathbb{R}^n \rightarrow \mathbb{R}$ be finite at a . A $\partial^+ p(a) \subseteq \mathbb{R}^n$ is called an *upper convexifactor* where $\partial^+ p(a)$ being a enclosed set of p at a if

$$p^\downarrow(m; \lambda) \leq \sup_{\xi \in \partial^+ p(m)} \langle \xi, \lambda \rangle \quad \forall \lambda \in \mathbb{R}^n.$$

Definition 4.2. Lower Convexifactor [19, 3]

A $\partial^- p(a) \subseteq \mathbb{R}^n$ is called a *lower convexifactor* where $\partial^- p(a)$ being a enclosed set of p at a if

$$p^\uparrow(m; \lambda) \geq \inf_{\xi \in \partial^- p(m)} \langle \xi, \lambda \rangle \quad \forall \lambda \in \mathbb{R}^n.$$

Definition 4.3. Convexifactor [19, 3]

The set $\partial p(a) \subseteq \mathbb{R}^n$ is called a *CF* of p at a if it is both an UC and a LC of p at a .

Several remarks are in order.

Remark 1. Unlike the Clarke subdifferential, a convexifactor need not be convex or compact. This relaxation is one of the main reasons convexifactors are useful in nonsmooth optimization, since they may provide sharper or more economical optimality conditions.

Remark 2. For a locally Lipschitz function p , both the Clarke generalized subdifferential $\partial^C p(a)$ and the Michel–Penot subdifferential are convexifactors of p at a ; see [19]. Hence the convexifactor framework extends the classical Clarke theory.

We now introduce two finer notions that play an important role in the chain rules and in the necessary optimality conditions.

Definition 4.4. USRCF [19, 3]] A $\partial^{us} p(a) \subseteq \mathbb{R}^n$ is called an *upper semi-regular CF* where $\partial^+ p(a)$ being a enclosed set of p at a if

$$p^\uparrow(a; v) \leq \sup_{\xi \in \partial^{us} p(a)} \langle \xi, v \rangle \quad \text{for all } v \in \mathbb{R}^n.$$

If equality holds for all v , then $\partial^{us} p(a)$ is called an *upper regular CF*

The relationship among these notions is summarized by

$$\text{upper regular CF} \implies \text{upper semi-regular CF} \implies \text{upper CF},$$

but no implication holds in the reverse direction, as the following examples show.

Example 4.1 (An Upper Convexifactor That Is Not Upper Semi-Regular). Let Q be the set of rationals and define p be a function from $\mathbb{R} \rightarrow \mathbb{R}$ by

$$p(t) = \begin{cases} t, & t \in Q, \\ 0, & t \notin Q. \end{cases}$$

At the point $t = 0$, one computes

$$p^\uparrow(0; v) = \max\{0, v\} \quad \text{and} \quad p^\downarrow(0; v) = \min\{0, v\}.$$

Consider $\partial p(0) := \{-1, 0\}$. For any $v \in \mathbb{R}$,

$$\sup_{\xi \in \partial p(0)} \langle \xi, v \rangle = \max\{-v, 0\}.$$

One checks that

$$p^\downarrow(0; v) \leq \sup_{\xi \in \partial p(0)} \langle \xi, v \rangle$$

upper convexifactor $\forall v$, so $\{-1, 0\}$ is an UCF. However, taking $v = 1$ gives

$$p^\uparrow(0; 1) = 1 \quad \text{while} \quad \sup_{\xi \in \partial p(0)} \langle \xi, 1 \rangle = 0 < 1,$$

so the upper semi-regularity condition fails. Hence $\{-1, 0\}$ is an UCF but not an upper semi-regular CF.

4.3.2 Compact CF Existence

The following result, which parallels Proposition 2.1 of Uderzo [25], guarantees the existence of compact convexifactors under mild boundedness hypotheses. The proof follows the same compactness argument used for upper convex approximations.

Theorem 4.1. Compact Convexifactors [25, 3]

Let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$ and let a belongs to $\mathbb{R}^n$ be a point where $p(a)$ is finite.

- (i) If $p_d^-(a; v)$ is bounded above for all v , then p admits a compact upper convexifactor at a .
- (ii) If $p_d^+(a; v)$ is bounded below for all v , then p admits a compact lower convexifactor at a .

Proof. We prove (i); the proof of (ii) is symmetric. Since $p_d^-(a; v)$ is bounded above and is positively homogeneous and subadditive in v , the function $v \mapsto p_d^-(a; v)$ being support function of nonempty compact convex set $C \subseteq \mathbb{R}^n$ (by theory of support functions; see [26]). The set C satisfies $p_d^-(a; v) \leq \sup_{\xi \in C} \langle \xi, v \rangle$ for all v , so C is a compact upper convexifactor. $\square$

4.3.3 Mean-Value Theorem for Convexifactors

The following MVT, due to Jeyakumar and Luc [19], is the key analytical tool underlying both chain rules in this chapter.

Theorem 4.2. Mean-Value Theorem for Convexifactors [19] Let $a, b \in \mathbb{R}^n$ and let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$ be continuous. Now, Assume at each $x \in (a, b)$, the sets $\partial^u p(x)$ and $\partial^\ell p(x)$ are, resp., upper and lower CF of p at x . Then $c \in (a, b)$ and a sequence $\{\xi^k\} \subseteq \text{co}(\partial^u p(c)) \cup \text{co}(\partial^\ell p(c))$ such that

$$p(b) - p(a) = \lim_{k \rightarrow \infty} \langle \xi^k, b - a \rangle.$$

If the convexifactor $\partial p(x)$ is bounded at each x , then the limit is attained: there exist $c \in (a, b)$ and $\xi \in \partial p(c)$ such that

$$p(b) - p(a) = \langle \xi, b - a \rangle.$$

4.3.4 Chain Rules for Convexifactors

We now establish the two chain rules that form the computational backbone of the convexifactor calculus.

Theorem 4.3. Chain Rule with Bounded Convexifactor [3]

Let $R: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable and therefore be continuous and let $q: \mathbb{R}^m \rightarrow \mathbb{R}$ be continuous. Let at each $y \in \mathbb{R}^m$ the function q admits a bounded convexifactor $\partial q(y)$, and that

the set-valued map $y \mapsto \partial q(y)$ is closed-valued. Then the composite function $q \circ R: \mathbb{R}^n \rightarrow \mathbb{R}$ admits an USRCF at each $a \in \mathbb{R}^n$, given by

$$\partial^{us}(q \circ R)(a) = JR(a)^\top \partial q(R(a)),$$

where $JR(a)$ denotes the Jacobian matrix of R at a .

Proof. Since q admits a bounded convexifactor at each y , it follows from Corollary 5.2 of [19] that q is locally Lipschitz on $\mathbb{R}^m$. Because R is continuously differentiable, the composition $r := q \circ R$ is locally Lipschitz on $\mathbb{R}^n$, so $r_d^+(a; v)$ is finite for all a, v .

Goal. We show that $r_d^+(a; v) \leq \sup_{\xi \in JR(a)^\top \partial q(R(a))} \langle \xi, v \rangle$ for all $v \in \mathbb{R}^n$.

Step 1: Extract a sequence realizing the lim sup. By definition of the upper Dini derivative a sequence $\varepsilon_i \downarrow 0$ where

$$r_d^+(a; v) = \lim_{i \rightarrow \infty} \frac{r(a + \varepsilon_i v) - r(a)}{\varepsilon_i}. \quad (1)$$

Step 2: Apply the Mean-Value Theorem. For each i , apply Theorem 4.2 (non-asymptotic form, which holds since ∂q is bounded) to q along the segment $[R(a), R(a + \varepsilon_i v)]$. This yields a point $c_i \in (R(a), R(a + \varepsilon_i v))$ and $\eta_i \in \partial q(c_i)$.

Step 3: Pass to the limit. As $i \rightarrow \infty$ we have $\varepsilon_i \rightarrow 0$, so $R(a + \varepsilon_i v) \rightarrow R(a)$ and thus $c_i \rightarrow R(a)$. Since $\partial q(c_i)$ is bounded, the sequence $\{\eta_i\}$ has a convergent subsequence (relabelled) with limit $\eta \in \mathbb{R}^m$. Because the set-valued map ∂q is closed-valued and $c_i \rightarrow R(a)$, we conclude $\eta \in \partial q(R(a))$. Dividing (2) by ε_i and using the differentiability of R :

$$\frac{R(a + \varepsilon_i v) - R(a)}{\varepsilon_i} \xrightarrow{i \rightarrow \infty} JR(a) v. \quad (3)$$

Combining (1)–(3):

$$r_d^+(m; v) = \langle \phi, JR(m) v \rangle = \langle JR(m)^\top \phi, v \rangle \leq \sup_{\xi \in JR(m)^\top \partial q(R(m))} \langle \xi, v \rangle.$$

Since $JR(m)^\top \partial q(R(m))$ is closed and the inequality holds for all v , it is an USRCF of r at m . $\square$

Remark 3. Set $JR(a)^\top \partial q(R(a))$ need not be convex even when $\partial q(R(a))$ is convex, because the transpose map $JR(a)^\top$ is linear. The result is nonetheless useful because the support function of $JR(a)^\top \partial q(R(a))$ equals that of its convex hull, so optimality conditions derived using this chain rule are equivalent to those one would obtain from the convex hull.

The next theorem extends the chain rule to the case where the convexifactor of q need not be bounded, at the cost of introducing an ε -perturbation.

Theorem 4.4. Chain Rule Without Boundedness [3] Let $R: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable and let q be a function from $\mathbb{R}^m \rightarrow \mathbb{R}$ be continuous with CF $\partial q(y)$ at each $y \in \mathbb{R}^m$. Assume that the set-valued map $y \mapsto \partial q(y)$ is upper semicontinuous. Then for every $\varepsilon > 0$, the composite function $q \circ R$ admits an USRCF at each $a \in \mathbb{R}^n$, given by

$$\partial_\varepsilon^{us}(q \circ R)(a) = (JR(a) + \varepsilon B_{\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)})^\top \text{co } \partial q(R(a)).$$

Proof. Set $A_\varepsilon := (JR(a) + \varepsilon B_{\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)})^\top \text{co } \partial q(R(a))$. We must show $r_d^+(a; v) \leq \sup_{\xi \in A_\varepsilon} \langle \xi, v \rangle$.

Step 1. Choose $\varepsilon_i \downarrow 0$ realizing $r_d^+(a; v)$ as in (1) of the proof of Theorem 4.3.

Step 2: Apply the asymptotic MVT. For each i , Theorem 4.2 yields $c_i \in (R(a), R(a + \varepsilon_i v))$ and a sequence $\{\eta_i^k\} \subseteq \text{co } \partial q(c_i)$ such that

$$q(R(a + \varepsilon_i v)) - q(R(a)) = \lim_{k \rightarrow \infty} \langle \eta_i^k, R(a + \varepsilon_i v) - R(a) \rangle.$$

Step 3: Use upper semicontinuity. Since ∂q is upper semicontinuous and $c_i \rightarrow R(a)$, for any fixed $\delta > 0$ and i sufficiently large we have $\text{co } \partial q(c_i) \subseteq \text{co } \partial q(R(a)) + \delta B_{\mathbb{R}^m}$. Hence for each k and large i : $\eta_i^k \in \text{co } \partial q(R(a)) + \delta B_{\mathbb{R}^m}$.

Step 4: Use differentiability of R . Since R is continuously differentiable, for i large enough: $\|[R(a + \varepsilon_i v) - R(a)]/\varepsilon_i - JR(a)v\| < \varepsilon$, which means the numerical Jacobian $[R(a + \varepsilon_i v) - R(a)]/\varepsilon_i$ lies within $(JR(a) + \varepsilon B)v$.

Step 5: Combine. For i sufficiently large and any k_i ,

$$\frac{\|R(a + \varepsilon_i v) - R(a)\|}{\varepsilon_i} \leq \sup\{\|x_i\| : x_i \in A(\varepsilon_i)\} + M\delta,$$

where M bounds the norms in $A(\varepsilon_i)$. As from above $\delta > 0$ is arbitrary, the result follows by letting $\delta \rightarrow 0$. $\square$

Remark 4. Theorems 4.3 and 4.4 cover the case where the outer function q is nonsmooth and the inner map R is smooth. The reverse case—where R is nonsmooth and q is smooth—requires the notion of an *approximate Jacobian* for R . The two cases are complementary and together make the convexifactor calculus applicable to multiple range of composite optimization problems.

4.4 Convexifactors and Generalized Convexity

4.4.1 ∂ -Pseudoconvexity

Now, we introduce a new notion of pseudoconvexity defined directly through the convexifactor, and study its relationship with the pseudomonotonicity of the convexifactor mapping. This

approach is different from—and in some respects more tractable than—the pseudoconvexity notion of Ortega and Rheinboldt [27].

Definition 4.5. ∂ -Pseudoconvex Function [3] Let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$ admit a convexfactor $\partial p(a)$ at each $a \in \mathbb{R}^n$. Then p is called ∂ -pseudoconvex if for all distinct $a, b \in \mathbb{R}^n$,

$$p(b) < p(a) \implies \langle \eta, b - a \rangle < 0 \quad \text{for all } \eta \in \partial p(a).$$

Remark 5. Every ∂ -pseudoconvex function is quasiconvex. Hence ∂ -pseudoconvex functions form a subclass of continuous quasiconvex functions. When $\partial p(a) = \partial^c p(a)$ (the Clarke sub-differential), ∂ -pseudoconvex functions are known in the literature as *nonsmooth pseudoconvex functions*.

Proposition 4.1. Local Minimum Implies Global Minimum [3] Let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$ be ∂ -pseudoconvex, and let $a^0 \in \mathbb{R}^n$ be a local minimizer of p . Now suppose that $\partial p(a^0)$ is bounded. Then a^0 is a global minimizer of p .

Proof. As a^0 is ome local minimizer and $\partial p(a^0)$ is bounded, Proposition 4.1 of [19] gives $0 \in \text{co } \partial p(a^0)$.

suppose that a^0 is not a global minimizer. Then $b \in \mathbb{R}^n$ with $p(b) < p(a^0)$. By ∂ -pseudoconvexity:

$$\langle \eta, b - a^0 \rangle < 0 \quad \text{for all } \eta \in \partial p(a^0).$$

This extends by linearity to all convex combinations:

$$\langle \eta, b - a^0 \rangle < 0 \quad \text{for all } \eta \in \text{co } \partial p(a^0).$$

But $0 \in \text{co } \partial p(a^0)$ gives $\langle 0, b - a^0 \rangle = 0$, a contradiction. Hence a^0 is a global minimizer. $\square$

4.4.2 Pseudomonotone Set-Valued Maps

Definition 4.6. Pseudomonotone and Quasimonotone Maps [3] A set-valued map $S: \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ is called *pseudomonotone* if $\forall \text{ distinct } m, n \in \mathbb{R}^n$,

$$\exists \eta^m \in S(m) : \langle \eta^m, n - a \rangle > 0 \implies \langle \eta^n, n - m \rangle > 0 \forall \xi^n \in S(n).$$

It is called *quasimonotone* if the conclusion is weakened to $\langle \eta^n, n - a \rangle \geq 0$.

The following lemma shows that pseudomonotonicity is stable under convex hull.

($\Leftarrow$) Since $S(a) \subseteq \text{co}(S(a))$, the pseudomonotonicity of $\text{co } S$ implies immediately that of S .

4.4.3 Pseudomonotonicity Implies ∂ -Pseudoconvexity

Theorem 4.5. Pseudomonotonicity Implies ∂ -Pseudoconvexity [3] Let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$ be some continuous function admitting CF $\partial p(a)$ at each a . If the map a maps to $\partial p(a)$ is pseudomonotone, then p is ∂ -pseudoconvex.

Proof. Suppose that p is not ∂ -pseudoconvex. Then $a, b \in \mathbb{R}^n$ with $p(b) < p(a)$ and some $\xi \in \partial p(a)$ such that $\langle \xi, b - a \rangle \geq 0$.

By the Mean-Value Theorem 4.2 applied to p on $[b, a]$ (using $p(b) - p(a) < 0$), there exists $c \in (b, a)$ and a sequence $\{\xi^k\} \subseteq \text{co}(\partial p(c))$ such that

$$\lim_{k \rightarrow \infty} \langle \xi^k, b - a \rangle = p(b) - p(a) < 0.$$

Hence for k large, $\langle \xi^k, a - b \rangle > 0$ with $\xi^k \in \text{co}(\partial p(c))$. Since $c \in (b, a)$, we have $\langle \xi^k, a - c \rangle > 0$.

By Lemma, $\text{co}(\partial p)$ is pseudomonotone (since ∂p is). Applying pseudomonotonicity at the pair (c, a) :

$$\langle \eta, a - c \rangle > 0 \quad \text{for all } \eta \in \text{co}(\partial p(a)).$$

In particular, $\langle \eta, a - b \rangle > 0$ for all $\zeta \in \text{co}(\partial p(a))$, which means $\langle \eta, b - a \rangle < 0 \forall \eta \in \text{co}(\partial p(a))$. But $\xi \in \partial p(a) \subseteq \text{co}(\partial p(a))$ and we assumed $\langle \xi, b - a \rangle \geq 0$ —a contradiction. $\square$

Remark 6. The converse of Theorem 4.5 holds when the convexifactor is upper regular at each point. This is shown next.

4.4.4 Quasiconvex Functions and Convexifactors

Theorem 4.6. [3] Let p be a function from $\mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous quasiconvex fnc admitting a convexifactor $\partial p(a)$ at each $a \in \mathbb{R}^n$. Suppose a dense subset $K \subseteq \mathbb{R}^n$ such that $\partial p(a)$ is UR $\forall a \in K$, and that for all $a \notin K$:

$$\partial p(a) \subseteq \overline{\{ \lim_{n \rightarrow \infty} \xi_n \mid \xi_n \in \partial p(a_n), a_n \in K, a_n \rightarrow a \}}. \quad (\star)$$

Then for all $a, b \in \mathbb{R}^n$:

$$p(b) < p(a) \implies \langle \xi, b - a \rangle \leq 0 \quad \text{for all } \xi \in \partial p(a).$$

Proof. Let $a, b \in \mathbb{R}^n$ with $p(b) < p(a)$. By MVT 4.2 there exists $c \in (b, a)$ and $\{\xi^k\} \subseteq \partial p(c)$ such that $\lim_k \langle \eta^k, b - a \rangle = p(b) - p(a) < 0$. Hence for k large, $\langle \eta^k, a - b \rangle > 0$, i.e. $\langle \eta^k, a - c \rangle > 0$ (since $c \in (b, a)$).

Since p is quasiconvex, the map ∂p is quasimonotone by Theorem 6.2 of [19]. Applying quasimonotonicity at the pair (c, a) gives $\langle \zeta, a - c \rangle \geq 0$ for all $\zeta \in \partial p(a)$, i.e. $\langle \zeta, b - a \rangle \leq 0$. $\square$

4.5 Convexifactors and Vector Optimization

4.5.1 Problem Setting and Weak Efficiency

We now apply the convexfactor framework to derive necessary optimality conditions for vector minimization problems. Consider the vector equation

$$F(a) = (p_1(a), \dots, p_k(a)) \quad \text{subject to } a \in C, \quad (\text{VOP})$$

where $C \subseteq \mathbb{R}^n$ is a closed set and each $p_j: \mathbb{R}^n \rightarrow \mathbb{R}$ is locally Lipschitz.

Definition 4.7. Weak Efficient Point At point $a^0 \in C$ of (VOP) if there does not exist $a \in C$ such that $p_j(a) < p_j(a^0)$ for all j .

Weak efficiency is a minimal requirement for optimality in vector optimization: it rules out points that are simultaneously dominated in every objective component. Stronger notions—Pareto efficiency, proper efficiency—impose further conditions but are not needed for the results in this chapter.

4.5.2 Necessary Conditions: Convex Feasible Set

Theorem 4.7. Necessary Condition, Convex Case [3] Consider (VOP) with C a closed convex set and each p_j locally Lipschitz. Suppose each p_j admits a bounded USRCF $\partial^{us} p_j(a^0)$ at a^0 . Then scalars $\lambda_j \geq 0$, not all zero, where let $a^0 \in C$ be a weak efficient point then

$$0 \in \sum_{j=1}^k \lambda_j \text{co } \partial^{us} p_j(a^0) + N(C, a^0).$$

Proof. Step 1: Directional derivative system has no solution. Since a^0 is weak efficient and each p_j is locally Lipschitz, we claim that the system

$$(p_1)_d^+(a^0; \eta) < 0, \dots, (p_k)_d^+(a^0; \eta) < 0 \quad (*)$$

has no solution $v \in T(C, a^0)$, where $T(C, x^0)$ is the Bouligand tangent cone.

Suppose (*) held for some $\eta \in S(C, m^0)$. Since C is convex, $S(C, m^0) = \text{cone}(C - m^0)$, so there exist scalars $\alpha_n > 0$ and points $a_n \in C$ with $v = \lim_n \alpha_n(m_n - m^0)$. By positive homogeneity and local Lipschitz continuity, for large z : $(p_j)_d^+(m^0; m_n - m^0) < 0, l = 1, \dots, k$. This gives some neighbourhood direction along which all p_j strictly decrease, violating weak efficiency.

Step 2: Translate via upper semi-regular convexifactors. Since each $\partial^{us} p_j(a^0)$ is upper semi-regular and bounded, $(p_j)_d^+(a^0; v) \leq \sup_{\xi \in \partial^{us} p_j(a^0)} \langle \xi, v \rangle$. Hence the system has no solution $v \in T(C, a^0)$.

Step 3: Separation. By a standard separation argument applied to the cones $\text{co } \partial^{us} p_j(a^0)$ and $T(C, a^0)$, there exist $\lambda_j \geq 0$, not all zero, such that

$$\sum_{j=1}^k \lambda_j \sup_{\xi \in \partial^{us} p_j(a^0)} \langle \xi, v \rangle \geq 0 \quad \text{for all } v \in T(C, a^0).$$

By the calculus of support functions:

$$\sup_{\xi \in \sum_j \lambda_j \text{co } \partial^{us} p_j(a^0)} \langle \xi, v \rangle \geq 0 \quad \text{for all } v \in T(C, a^0).$$

This is equivalent to $0 \in \sum_j \lambda_j \text{co } \partial^{us} p_j(a^0) + N(C, a^0)$. □

Remark 7. Theorem 4.7 can yield sharper Lagrange multiplier conditions than the classical Clarke-based theory, because the convex hull of a bounded upper semi-regular convexifactor may be strictly smaller than the Clarke subdifferential (see Remark 4.1 of [19]).

4.6 Conclusion

The chapter has given a complete self-contained treatment of the. convexifactor model of Dutta and Chandra [3]. The main themes: contributions of the chapter are as follows.

Calculus of convexifactors. We presented upper convexifactors, lower convexifactors, upper semi-regular convexifactors, and regular convexifactors upper, and of which, by way of examples, there is a strict hierarchy. The All further theorems rely on Mean-Value Theorem (Theorem 4.2). analysis. These are the two chain rules (Theorems 4.3 and 4.4) demonstrate the behaviour of convexifactors under. composition satisfying a smooth map, with both the boundedness assumption and without. These are necessary to treat the composite nonsmooth functions. that are the result of the bilevel reformulation in the following chapters.

Generalized convexity. The concept of the partial-pseudoconvexity (Definition 4.5) offers a nonsmooth version of classical pseudoconvexity developed. directly in convexifactor. It has the following important properties: (i) all functions that are quasiconvex are partial-pseudoconvex.

Vector optimization. Theorems 4.7, ??, and ?? deduce what can be little known. efficiency in minimization of vectors. The conditions are of the form of the generalized KKT-type inclusions which include convex hulls of convexifactors. and the ordinary cone. One peculiarity is that upper semi-regularity can be slackened to at most one objective component (Remark 4.2). of [3]) which provides some flexibility not seen in the Clarke subdifferential framework.

Role in the dissertation This chapter is the analytical toolkit based on its results. throughout the remainder of the dissertation. Invocation of the chain rules will be invoked. The nonsmooth value function is dealt with in Chapter 5, gap function that occurs due to the bilevel reformulation. The generalized convexity ideas- ∂ -pseudoconvexity and ∂ -quasiconvexity, the latter, which was presented in Chapter 4–will look like. sufficient optimality conditions and duality theorems of. Chapters 4 and 6. In this sense, the convexifactor calculus developed here is not merely background material but the active mathematical language in which the subsequent results are expressed.

Chapter 5

Unified Framework for Optimality and Duality in Bilevel Multiobjective Programming

5.1 Introduction

The chapter develops a combined analytical structure for bilevel programming in two settings. The first concerns a scalar upper-level objective and uses the follower's value function to convert the bilevel model into an equivalent single-level programme. The second replaces the scalar objective with a vector-valued one, encountering an analytical obstacle—the failure of the Mangasarian–Fromovitz condition at feasible points—that requires a different reformulation strategy.

For the scalar bilevel problem (BP), the treatment follows Suneja and Kohli [7]. Five generalised convexity classes for bifunctions, defined through convexifactors (Definition 5.4), are introduced and shown to be mutually distinct. These classes support sufficient optimality conditions (Theorems 5.2 and 5.3) and a complete Wolfe-type and Mond–Weir-type duality theory (Theorems 5.4–5.7).

For the multiobjective bilevel problem (MBP), the treatment follows Gadhi and Dempe [4]. The Ξ -reformulation (Section 5.9) encodes the lower-level optimality condition $b \in \Lambda(a)$ as the scalar inequality $\Xi(a, b) \leq 0$ through the Hiriart-Urruty gauge function Δ_T . Unlike the value-function constraint, the Ξ -constraint can satisfy standard constraint qualifications, enabling the derivation of KKT-type necessary conditions (Theorems 5.8 and 5.9) without ad hoc regularity assumptions.

Both parts draw on the convexifactor calculus of Chapter 3 [3, 19]: the chain rules, mean-value theorem, and the hierarchy of lower level, upper semi-regular convexifactors or upper level all appear in the proofs. The shared definitions needed for both analyses are collected in Section 5.2, so that each later section can refer back rather than repeat them.

5.2 Analytical Preliminaries

All function spaces are finite-dimensional real Euclidean spaces. The inner product and norm on $\mathbb{R}^n$ are $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$; $B_{\mathbb{R}^n}$ is the closed unit ball. Conventions for $\text{co } A$, $\overline{\text{co}} A$, and related set

operations follow Chapter 2 (Section 2.2).

Preliminary 5.1 (Locally Lipschitz Functions [1]). A map $p: \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \rightarrow \mathbb{R}$ is *locally Lipschitz* at $(\bar{a}, \bar{b})$ when a constant $K \geq 0$ and a radius $r > 0$ exist such that

$$|p(a_1, b_1) - p(a_2, b_2)| \leq K \|(a_1, b_1) - (a_2, b_2)\|$$

for all $(a_1, b_1), (a_2, b_2) \in B((\bar{a}, \bar{b}), r)$.

Preliminary 5.2 (Dini Directional Derivatives [3]). Let p be finite at $(\bar{a}, \bar{b})$ and let $d = (d_1, d_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$. The *upper* and *lower Dini directional derivatives* of p at $(\bar{a}, \bar{b})$ in direction d are, respectively,

$$p^{[+]}((\bar{a}, \bar{b}); d) := \limsup_{t \downarrow 0} \frac{p((\bar{a}, \bar{b}) + td) - p(\bar{a}, \bar{b})}{t},$$

$$p^{[-]}((\bar{a}, \bar{b}); d) := \liminf_{t \downarrow 0} \frac{p((\bar{a}, \bar{b}) + td) - p(\bar{a}, \bar{b})}{t}.$$

One always has $p^{[-]} \leq p^{[+]}$; both are finite when p is locally Lipschitz.

Preliminary 5.3 (Clarke Subdifferential [1]). For p locally Lipschitz at $(\bar{a}, \bar{b})$, the *Clarke subdifferential* is

$$\partial^c p(\bar{a}, \bar{b}) := \text{co} \left\{ \lim_{k \rightarrow \infty} \nabla p(a_k, b_k) \mid (a_k, b_k) \rightarrow (\bar{a}, \bar{b}), p \text{ differentiable at each } (a_k, b_k) \right\}.$$

This set is nonempty, convex, compact. The associated *Clarke directional derivative* is $p^\circ((\bar{a}, \bar{b}); d) := \max_{\zeta \in \partial^c p(\bar{a}, \bar{b})} \langle \zeta, d \rangle$.

Preliminary 5.4 (Convexifactor of a Bifunction [7, 3]). Let p be finite at $(\bar{a}, \bar{b})$.

1. A closed set $\partial^u p(\bar{a}, \bar{b})$ is an *UCFr* if $p^{[-]}((\bar{a}, \bar{b}); d) \leq \sup_{\zeta \in \partial^u p(\bar{a}, \bar{b})} \langle \zeta, d \rangle$ for all d .
2. A closed set $\partial^l p(\bar{a}, \bar{b})$ is a *LCF* if $p^{[+]}((\bar{a}, \bar{b}); d) \geq \inf_{\zeta \in \partial^l p(\bar{a}, \bar{b})} \langle \zeta, d \rangle$ for all d .
3. A closed set $\partial^* p(\bar{a}, \bar{b})$ is a *CF* when it is simultaneously a UCF and an LCF.

Remark 8. Unlike the Clarke subdifferential, a convexifactor carries no requirement of convexity or compactness. For locally Lipschitz p , the set $\partial^c p$ is always a convexifactor; however, $\text{co}(\partial^* p)$ may be strictly smaller than $\partial^c p$, which is why convexifactors can yield sharper multiplier conditions [3, 19].

Preliminary 5.5 (Hiriart-Urruty Gauge Function [6]). For $\emptyset \neq T \subsetneq \mathbb{R}^n$, the *Hiriart-Urruty gauge* $\Delta_T: \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by

$$\Delta_T(w) := \begin{cases} -d(w, \mathbb{R}^n \setminus T) & w \in T, \\ d(w, T) & w \notin T, \end{cases}$$

where $d(w, A) := \inf_{a \in A} \|w - a\|$. When T is a closed convex cone, Δ_T is convex, positively homogeneous, 1-Lipschitz, and order-decreasing with respect to the cone order. In particular, $T = \{w : \Delta_T(w) \leq 0\}$ and $\text{int } T = \{w : \Delta_T(w) < 0\}$.

Preliminary 5.6 (Contingent Cone [2]). For $E \subseteq \mathbb{R}^n$ and $\bar{w} \in E$, the *contingent cone* to E at $\bar{w}$ is

$$\mathcal{K}(E, \bar{w}) := \{d \in \mathbb{R}^n : \exists t_k \downarrow 0, d_k \rightarrow d \text{ with } \bar{w} + t_k d_k \in E \forall k\}.$$

5.3 Scalar Bilevel Problem

5.3.1 Problem Statement and Value-Function Reformulation

The problem is:

$$_{a,b} p(a, b) \quad \text{subject to} \quad q_j(a, b) \leq 0 \ (j \in \mathcal{J}), \quad b \in \Sigma(a), \quad (\text{BP})$$

where $\Sigma(a)$ collects all optimal solutions of the lower-level equation

$$_b r(a, b) \quad \text{subject to} \quad \kappa_i(a, b) \leq 0 \ (i \in \mathcal{I}). \quad (\text{LL}_a)$$

The functions p, r, q_j, κ_i are all locally Lipschitz. The lower-level feasible set for fixed a is $\mathcal{Y}(a) := \{b : \kappa_i(a, b) \leq 0 \ \forall i\}$, and the optimal value function is

$$\mathcal{V}(a) := \inf\{r(a, b) : b \in \mathcal{Y}(a)\}. \quad (5.1)$$

Under this assumption, (BP) is equivalent to the single-level programme:

$$\min_{a,b} p(a, b) \quad \text{subject to} \quad r(a, b) - \mathcal{V}(a) \leq 0, \quad \kappa_i(a, b) \leq 0, \quad q_j(a, b) \leq 0. \quad (\text{RBP})$$

Lemma 5.1 (Equivalence of (BP) and (RBP) [7]). A pair $(\bar{a}, \bar{b})$ is an optimal solution of (BP) if and only if it is an optimal solution of (RBP).

Proof. If $(\bar{a}, \bar{b})$ solves (BP), then $\bar{b} \in \Sigma(\bar{a})$ implies $r(\bar{a}, \bar{b}) = \mathcal{V}(\bar{a})$, so the first constraint of (RBP) holds with equality; the remaining constraints are inherited. Conversely, if $(\bar{a}, \bar{b})$ solves (RBP), then $r(\bar{a}, \bar{b}) \leq \mathcal{V}(\bar{a})$ together with $\bar{b} \in \mathcal{Y}(\bar{a})$ gives $r(\bar{a}, \bar{b}) \geq \mathcal{V}(\bar{a})$, so $\bar{b} \in \Sigma(\bar{a})$; both problems share the objective p . $\square$

Remark 9 (Existence Conditions [7]). The following conditions guarantee that (BP) has an optimal solution:

1. All data functions are lower semicontinuous.
2. The value function $\mathcal{V}$ is upper semicontinuous.

3. For some constant c exceeding the infimal value of (RBP), the set $\{(a, b) : q(a, b) \leq 0, \kappa(a, b) \leq 0, p(a, b) \leq c\}$ is nonempty and bounded.

These conditions are assumed throughout the scalar bilevel part.

5.4 Generalised Convexity Classes for Bilevel Analysis

The following five definitions extend the convexifactor framework of Dutta and Chandra [3] to the bilevel setting [7]. Throughout, p is a bifunction admitting a convexifactor $\partial^* p(\bar{a}, \bar{b})$.

Definition 5.1 (∂^* -Convex Bifunction [7]). The bifunction p is ∂^* -convex at $(\bar{a}, \bar{b})$ if, for every (a, b) and every $\zeta \in \partial^* p(\bar{a}, \bar{b})$,

$$p(a, b) \geq p(\bar{a}, \bar{b}) + \langle \zeta, (a, b) - (\bar{a}, \bar{b}) \rangle. \quad (5.2)$$

When strict inequality holds for $(a, b) \neq (\bar{a}, \bar{b})$, the function is *strictly ∂^* -convex*.

Definition 5.2 (∂^* -Pseudoconvex Bifunction [7]). The bifunction p is ∂^* -pseudoconvex at $(\bar{m}, \bar{n})$ if, for every (m, n) and every $\zeta \in \partial^* p(\bar{m}, \bar{n})$,

$$p(m, n) < p(\bar{m}, \bar{n}) \implies \langle \zeta, (m, n) - (\bar{m}, \bar{n}) \rangle < 0.$$

If the premise $p(a, b) < p(\bar{m}, \bar{n})$ is replaced by $p(m, n) \leq p(\bar{m}, \bar{n})$ for $(m, n) \neq (\bar{m}, \bar{n})$, the function is *strictly ∂^* -pseudoconvex*.

Definition 5.3 (∂^* -Quasiconvex Bifunction [7]). The bifunction p is ∂^* -quasiconvex at $(\bar{m}, \bar{n})$ if, for every (m, n) and every $\zeta \in \partial^* p(\bar{m}, \bar{n})$,

$$p(m, n) \leq p(\bar{m}, \bar{n}) \implies \langle \zeta, (m, n) - (\bar{m}, \bar{n}) \rangle \leq 0.$$

5.4.1 Logical Hierarchy of the Five Classes

Remark 10 ([7]). ∂^* -convexity implies ∂^* -pseudoconvexity, but not conversely. Every strictly ∂^* -pseudoconvex function is ∂^* -pseudoconvex. ∂^* -pseudoconvexity and ∂^* -quasiconvexity are independent. ∂^* -convexity implies ∂^* -quasiconvexity.

Example 5.1 (∂^* -Pseudoconvex but not ∂^* -Convex [7]). Define $p(a, b) := \sqrt{a+b}$ for $a, b > 0$, $\sqrt{a-b}$ for $a > 0, b \leq 0$, $\sqrt{-a+b}$ for $a \leq 0, b > 0$, and $-\sqrt{-a+b}$ for $a \leq 0, b \leq 0$, with $\partial^* p(0, 0) = \{(\zeta_1, \zeta_2) : \zeta_1 \geq 1, \zeta_2 \geq 1\}$. Whenever $p(a, b) < 0 = p(0, 0)$ and $\zeta \in \partial^* p(0, 0)$, one obtains $\langle \zeta, (a, b) \rangle < 0$, confirming ∂^* -pseudoconvexity. Yet for $(a, b) = (1, -2)$ and $\zeta = (6, 1)$, $p(1, -2) = 1 < 0 + \langle (6, 1), (1, -2) \rangle = 4$, violating (5.2), so ∂^* -convexity fails.

Example 5.2 (∂^* -Quasiconvex but not ∂^* -Pseudoconvex [7]). Define $p(a, b) := \sqrt{a^2 b}$ for $a \geq 0, b \geq 0$; $\sqrt{-ab}$ for $a \geq 0, b < 0$; $-\sqrt{-ab^2}$ for $a < 0, b \geq 0$; $\sqrt{-a^2 b}$ for $a < 0, b < 0$, with

$\partial^* p(0, 0) = \{(\zeta_1, 0) : \zeta_1 \geq 0\}$. At $(-1, 2)$: $p(-1, 2) = 0 = p(0, 0)$ and $\langle (\zeta_1, 0), (-1, 2) \rangle = -\zeta_1 \leq 0$, so ∂^* -quasiconvexity holds. Taking $\zeta = (0, 0)$, however, $p(-1, 2) = 0 \leq p(0, 0)$ but $\langle (0, 0), (-1, 2) \rangle = 0$, failing the strict inequality of ∂^* -pseudoconvexity.

Example 5.3 (Strictly ∂^* -Convex Quadratic). For $p(a, b) = a^2 + b^2$, the convexifactor at any $(\bar{a}, \bar{b})$ is the singleton $\{(2\bar{a}, 2\bar{b})\}$. Since $p(a, b) - p(\bar{a}, \bar{b}) - \langle (2\bar{a}, 2\bar{b}), (a - \bar{a}, b - \bar{b}) \rangle = (a - \bar{a})^2 + (b - \bar{b})^2 > 0$ for $(a, b) \neq (\bar{a}, \bar{b})$, the function is strictly ∂^* -convex at every point.

5.5 Regularity and Necessary Optimality Conditions

Definition 5.4 (Regularity [7, 10]). Problem (BP) is *regular* at a feasible point $(\bar{a}, \bar{b})$ if there exist a neighbourhood $\mathcal{U}$ of $(\bar{a}, \bar{b})$ and constants $\delta, \theta > 0$ such that for all $(\beta, \sigma) \in \mathbb{R}_+^{m_1} \times \mathbb{R}_+^{m_2}$, $(a, b) \in \mathcal{U}$, and appropriate subdifferential selections $\zeta_\kappa, \zeta_q, \zeta_r, \zeta_{\mathcal{V}}$, there exists $d \in \delta B_{\mathbb{R}^{n_1}}$ with

$$\beta \cdot \kappa(m, n) + \sigma \cdot q(m, n) + \langle \zeta_\kappa + \zeta_q + \zeta_r - \zeta_{\mathcal{V}}, d \rangle \geq \theta.$$

This condition guarantees a non-degenerate Lagrange multiplier for p in the optimality conditions below.

Define the *active constraints* at $(\bar{a}, \bar{b})$:

$$\mathcal{A}_q := \{j \in \mathcal{J} : q_j(\bar{a}, \bar{b}) = 0\}, \quad \mathcal{A}_\kappa := \{i \in \mathcal{I} : \kappa_i(\bar{a}, \bar{b}) = 0\},$$

and the *lower-level active set* $\mathcal{R}(\bar{a}) := \{b : \kappa(\bar{a}, b) \leq 0, r(\bar{a}, b) = \mathcal{V}(\bar{a})\}$.

Theorem 5.1 (Necessary Optimality Conditions [7]). Let $(\bar{a}, \bar{b})$ be optimal for (BP). Suppose the convexifactor maps are upper semicontinuous and bounded at $(\bar{a}, \bar{b})$. Then scalars $\lambda_1, \lambda_2, \alpha \geq 0$ with $\lambda_1 + \lambda_2 = 1$ and multipliers $\beta_i \geq 0, \sigma_j \geq 0$ with $\|(\beta, \sigma, \alpha)\| = 1$ exist, satisfying

$$\sum_i \beta_i \kappa_i(\bar{a}, \bar{b}) = 0, \quad \sum_j \sigma_j q_j(\bar{a}, \bar{b}) = 0, \tag{5.3}$$

$$\begin{aligned} (0, 0) \in & \lambda_1 \text{co } \partial^* p(\bar{a}, \bar{b}) + \lambda_2 \alpha \text{co } \partial^* r(\bar{a}, \bar{b}) + \lambda_2 \sum_i \beta_i \text{co } \partial^* \kappa_i(\bar{a}, \bar{b}) \\ & + \lambda_2 \sum_j \sigma_j \text{co } \partial^* q_j(\bar{a}, \bar{b}) - \lambda_2 \alpha (\partial^* \mathcal{V}(\bar{a}) \times \{0\}), \end{aligned} \tag{5.4}$$

where $\partial^* \mathcal{V}(\bar{a}) := \text{co}\{\partial^* r(\cdot, b)(\bar{a}) : b \in \mathcal{R}(\bar{a})\}$. Moreover, if (BP) is regular at $(\bar{a}, \bar{b})$, then $\lambda_1 > 0$.

5.6 Sufficient Optimality Conditions

Theorem 5.2 (Sufficient Conditions: ∂^* -Convex Case [7]). Let $(\bar{a}, \bar{b})$ be feasible for (BP). If $p, r - \mathcal{V}, q_j$ ($j \in \mathcal{J}$), and κ_i ($i \in \mathcal{I}$) are all ∂^* -convex at $(\bar{a}, \bar{b})$, and if scalars $\alpha \geq 0, \beta_i \geq 0$,

$\sigma_j \geq 0$ satisfy

$$(0, 0) \in \text{co } \partial^* p(\bar{a}, \bar{b}) + \alpha \text{co}(\partial^* r(\bar{a}, \bar{b}) - \partial^* \mathcal{V}(\bar{a}) \times \{0\}) + \sum_i \beta_i \text{co } \partial^* \kappa_i(\bar{a}, \bar{b}) + \sum_j \sigma_j \text{co } \partial^* q_j(\bar{a}, \bar{b}) \quad (5.5)$$

together with (5.3), then $(\bar{a}, \bar{b})$ is optimal for (BP).

Proof. Assume for contradiction that a feasible (a, b) for (RBP) satisfies $p(\bar{a}, \bar{b}) > p(a, b)$.

Step 1. ∂^* -convexity of p gives, for all $\eta_p \in \text{co } \partial^* p(\bar{a}, \bar{b})$,

$$p(a, b) \geq p(\bar{a}, \bar{b}) + \langle \eta_p, (a, b) - (\bar{a}, \bar{b}) \rangle. \quad (5.6)$$

Step 2. Condition (5.5) provides elements $\eta_p, \eta_r, \eta_{\kappa_i}, \eta_{q_j}$ from the respective convex hulls with $\eta_p + \alpha \eta_r + \sum_i \beta_i \eta_{\kappa_i} + \sum_j \sigma_j \eta_{q_j} = 0$.

Step 3. ∂^* -convexity of q_j and feasibility $q_j(a, b) \leq 0$ give $\langle \sum_j \sigma_j \eta_{q_j}, (a, b) - (\bar{a}, \bar{b}) \rangle \leq \sum_j \sigma_j (q_j(a, b) - q_j(\bar{a}, \bar{b})) \leq 0$ by (5.3). Identical steps for κ_i and $r - \mathcal{V}$ yield three further non-positive quantities.

Step 4. Setting η_p from Step 2 in (5.6) and substituting the zero-sum relation yields $p(a, b) \geq p(\bar{a}, \bar{b})$, a contradiction. $\square$

Theorem 5.3 (Sufficient Conditions: ∂^* -Pseudo/Quasi Case [7]). Let $(\bar{a}, \bar{b})$ be feasible for (BP). If p is ∂^* -pseudoconvex and $r - \mathcal{V}$, κ_i ($i \in \mathcal{A}_\kappa$), q_j ($j \in \mathcal{A}_q$) are ∂^* -quasiconvex at $(\bar{a}, \bar{b})$, and if (5.5) and (5.3) hold, then $(\bar{a}, \bar{b})$ is optimal for (BP).

Proof. Suppose a feasible (a, b) has $p(\bar{a}, \bar{b}) > p(a, b)$. ∂^* -pseudoconvexity gives $\langle \eta_p, (a, b) - (\bar{a}, \bar{b}) \rangle < 0$. ∂^* -quasiconvexity of the constraint functions, feasibility, and complementary slackness give three non-positive inner products for the constraint terms. Adding all four and using the zero-sum relation from (5.5) yields a strict negative, contradicting the zero sum. $\square$

Example 5.4 ([7]). Let $p(a, b) = a^2 + b$ (for $b \geq 0$), lower-level $r(a, b) = a^2 - b$ (for $a \geq 0, b \leq 0$), $\kappa_1(a, b) = b \leq 0$, $\kappa_2(a, b) = -b - 1 \leq 0$, and $q(a, b) = a^2 + b$. The value function is $\mathcal{V}(a) = a^2$ for $a \geq 0$. At $(\bar{a}, \bar{b}) = (0, 0)$: all functions are ∂^* -convex there, and with $\alpha = \frac{1}{4}$, $\beta_1 = \frac{1}{16}$, $\beta_2 = \frac{3}{4}$, $\sigma = \frac{1}{4}$, condition (5.5) and slackness (5.3) are both satisfied. Theorem 5.2 therefore confirms that $(0, 0)$ is optimal, consistent with $p(a, \Sigma(a)) = a^2$ minimised at $a = 0$.

5.7 Duality for the Scalar Bilevel Problem

Let $\mathbf{m} := (\alpha, \beta, \sigma)$ with $\alpha \geq 0, \beta_i \geq 0, \sigma_j \geq 0$.

5.7.1 Wolfe-Type Dual

Definition 5.5 (Wolfe-Type Dual (WD) [7]). The *Wolfe-type dual* of (BP) is

$$\max_{(u,v), \mathbf{m}} \mathcal{L}((u, v), \mathbf{m}), \quad (\text{WD})$$

subject to the stationarity inclusion

$$(0, 0) \in \text{co } \partial^* p(u, v) + \alpha \text{co}(\partial^* r(u, v) - \partial^* \mathcal{V}(u) \times \{0\}) + \sum_i \beta_i \text{co } \partial^* \kappa_i(u, v) + \sum_j \sigma_j \text{co } \partial^* q_j(u, v), \quad (5.7)$$

where the *Lagrangian objective* is

$$\mathcal{L}((u, v), \mathbf{m}) := p(u, v) + \alpha(r(u, v) - \mathcal{V}(u)) + \sum_i \beta_i \kappa_i(u, v) + \sum_j \sigma_j q_j(u, v). \quad (5.8)$$

Theorem 5.4 (Weak Duality: (WD) [7]). Suppose that for (BP) let (a, b) be feasible and $((u, v), \mathbf{m})$ be feasible for (WD). If $p, r - \mathcal{V}, q_j, \kappa_i$ are all ∂^* -convex at (u, v) , then $p(a, b) \geq \mathcal{L}((u, v), \mathbf{m})$.

Proof. ∂^* -convexity of p at (u, v) gives, for $\eta_p \in \text{co } \partial^* p(u, v)$, $p(a, b) \geq p(u, v) + \langle \eta_p, (a, b) - (u, v) \rangle$. Extract elements $\eta_p, \eta_r, \eta_{\kappa_i}, \eta_{q_j}$ from (5.7) summing to zero. ∂^* -convexity of q_j and feasibility $q_j(a, b) \leq 0$ yield $\langle \sum_j \sigma_j \eta_{q_j}, (a, b) - (u, v) \rangle \leq -\sum_j \sigma_j q_j(u, v)$. Similar bounds hold for κ_i and $r - \mathcal{V}$ (using $r(a, b) - \mathcal{V}(a) \leq 0$). Substituting into the bound for p and collecting gives $p(a, b) \geq \mathcal{L}((u, v), \mathbf{m})$. $\square$

Theorem 5.5 (Strong Duality: (WD) [7]). Let $(\bar{a}, \bar{b})$ be optimal for (BP) and suppose the hypotheses of Theorem 5.1 hold with (BP) regular at $(\bar{a}, \bar{b})$. Then $\bar{\mathbf{m}} = (\bar{\alpha}, \bar{\beta}, \bar{\sigma}) \geq 0$ exists such that $((\bar{a}, \bar{b}), \bar{\mathbf{m}})$ is feasible for (WD) and $p(\bar{a}, \bar{b}) = \mathcal{L}((\bar{a}, \bar{b}), \bar{\mathbf{m}})$. If Theorem 5.4 holds at every dual-feasible point, this is the optimal solution of (WD).

Proof. From Theorem 5.1 with $\lambda_1 > 0$, set $\bar{\alpha} := \lambda_2 \alpha / \lambda_1$, $\bar{\beta}_i := \beta_i \lambda_2 / \lambda_1$, $\bar{\sigma}_j := \sigma_j \lambda_2 / \lambda_1$. Dividing (5.4) by λ_1 gives (5.7). By Lemma 5.1, $r(\bar{a}, \bar{b}) - \mathcal{V}(\bar{a}) = 0$, and (5.3) kills the remaining Lagrangian terms, so $\mathcal{L}((\bar{a}, \bar{b}), \bar{\mathbf{m}}) = p(\bar{a}, \bar{b})$. Optimality follows from Theorem 5.4. $\square$

5.7.2 Mond–Weir-Type Dual

Definition 5.6 (Mond–Weir-Type Dual (MWD) [7]). The (MWD) of (BP) is

$$\max_{(u,v), \mathbf{m}} p(u, v) \quad (\text{MWD})$$

subject to (5.7) and the sign conditions

$$r(u, v) - \mathcal{V}(u) \geq 0, \quad \beta_i \kappa_i(u, v) \geq 0, \quad \sigma_j q_j(u, v) \geq 0, \quad \alpha, \beta_i, \sigma_j \geq 0. \quad (5.9)$$

Remark 11. The Mond–Weir dual maximises the primal objective p directly rather than the augmented Lagrangian $\mathcal{L}$. Its weak duality theorem requires only ∂^* -pseudoconvexity of p and ∂^* -quasiconvexity of the constraints—strictly weaker than the ∂^* -convexity demanded for the Wolfe dual [7].

Theorem 5.6 (Weak Duality: (MWD) [7]). Suppose that for (BP) let (a, b) be feasible and $((u, v), \mathbf{m})$ be feasible for (MWD). If p is ∂^* -pseudoconvex and $r - \mathcal{V}$, $\beta_i \kappa_i$, $\sigma_j q_j$ are ∂^* -quasiconvex at (u, v) , then $p(a, b) \geq p(u, v)$.

Proof. Suppose $p(a, b) < p(u, v)$. ∂^* -pseudoconvexity of p at (u, v) gives $\langle \eta_p, (a, b) - (u, v) \rangle < 0$ for $\eta_p \in \partial^* p(u, v)$. From (5.9), $r(u, v) - \mathcal{V}(u) \geq 0$ while feasibility gives $r(a, b) - \mathcal{V}(a) \leq 0$; ∂^* -quasiconvexity of $r - \mathcal{V}$ gives $\langle \alpha \eta_r, (a, b) - (u, v) \rangle \leq 0$. Identical arguments for κ_i and q_j provide two further non-positive inner products. Summing all four with the zero-sum relation from (5.7) yields a strict negative—a contradiction. $\square$

Theorem 5.7 (Strong Duality: (MWD) [7]). Under the hypotheses of Theorem 5.5, the point $((\bar{a}, \bar{b}), \bar{\mathbf{m}})$ constructed there is feasible for (MWD), the objective values coincide, and it is optimal for (MWD) whenever Theorem 5.6 applies at every dual-feasible point.

Proof. Complementary slackness gives $\bar{\beta}_i \kappa_i(\bar{a}, \bar{b}) = 0 \geq 0$, $\bar{\sigma}_j q_j(\bar{a}, \bar{b}) = 0 \geq 0$, and $r(\bar{a}, \bar{b}) - \mathcal{V}(\bar{a}) = 0 \geq 0$, establishing (5.9). Both problems attain value $p(\bar{a}, \bar{b})$. Optimality follows from Theorem 5.6. $\square$

5.8 The Multiobjective Bilevel Problem

The scalar objective p is now replaced by a vector map $\Phi = (\Phi_1, \dots, \Phi_n)$. The lower-level objective is renamed ℓ to distinguish it from the scalar setting. The *multiobjective bilevel problem* is:

$$\begin{aligned} & \mathbb{R}_+^n\text{-min}_{(a,b)} \Phi(a, b) \\ & \text{s.t. } H_j(a, b) \leq 0, \quad j \in \mathcal{M}, \quad b \in \Lambda(a), \end{aligned} \tag{MBP}$$

where $\Lambda(a)$ is known as the solution set of the lower-level equation

$$\min_b \ell(a, b) \quad \text{s.t. } \rho_s(a, b) \leq 0, \quad s \in \mathcal{S}. \tag{LL_a^m}$$

All functions $\Phi_i, H_j, \ell, \rho_s$ are locally Lipschitz [4]. Define: lower-level feasible set $\mathcal{P}(a) := \{b : \rho_s(a, b) \leq 0 \forall s\}$; lower-level value function $\mathcal{U}(a) := \inf\{\ell(a, b) : b \in \mathcal{P}(a)\}$; overall feasible set $\mathcal{F} := \{(a, b) : H_j(a, b) \leq 0 \forall j, b \in \Lambda(a)\}$.

Definition 5.7 (Locally Weak Efficient Solution [4, 8]). Some pair $(\bar{a}, \bar{b}) \in \mathcal{F}$ is a *local weak efficient solution* of (MBP) if a neighbourhood $\mathcal{N}$ of $(\bar{a}, \bar{b})$ exists such that $\Phi(a, b) - \Phi(\bar{a}, \bar{b}) \notin -\text{int } \mathbb{R}_+^n$ for all $(a, b) \in \mathcal{N} \cap \mathcal{F}$.

Remark 12. Applying the value-function constraint $\ell(a, b) - \mathcal{U}(a) \leq 0$ in the multiobjective context causes the Mangasarian–Fromovitz condition to fail at every feasible point. The Ξ -reformulation introduced below circumvents this obstruction [4, 32].

5.9 The Ξ -Reformulation of (MBP)

5.9.1 Compact Reference Set

Let $(\bar{a}, \bar{b})$ be a local weak efficient solution of (MBP). Assume the map $\mathcal{P}(\cdot)$ is uniformly bounded around $\bar{a}$, and let $\mathcal{D}_a$ be a neighbourhood of $\bar{a}$ with this property. Define

$$\Gamma := \text{cl}\left(\bigcup_{a \in \mathcal{D}_a} \mathcal{P}(a)\right) + B_{\mathbb{R}^{n_2}}. \quad (5.10)$$

Then Γ is compact and $\mathcal{P}(a) \subseteq \Gamma$ for all $a \in \mathcal{D}_a$ [4].

5.9.2 The Ξ -Function

For $(a, b) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ and $c \in \Gamma$, set

$$\xi(m, n, o) := \min\{\ell(m, n) - \ell(m, o), -\Delta_{(-\mathbb{R}_+^q)}(\rho_1(m, o), \dots, \rho_q(m, o))\}, \quad (5.11)$$

and

$$\Xi(a, b) := \max_{c \in \Gamma} \xi(m, n, o). \quad (5.12)$$

Since Γ is compact and ℓ, ρ_s are locally Lipschitz, the map $c \mapsto \xi(m, n, o)$ is continuous for each fixed (m, n) ; Ξ is therefore well defined and locally Lipschitz [4, 1].

5.9.3 Key Equivalence Lemmas

Lemma 5.2 (Non-existence of Strict Sub-optimality [4]). For $a \in \mathcal{D}_a$, $\{(a, b) : b \in \mathcal{P}(a), \ell(a, b) < \mathcal{U}(a)\} = \{(a, b) : b \in \mathcal{P}(a), \Xi(a, b) < 0\} = \emptyset$.

Proof. If $b \in \mathcal{P}(a)$ and $\Xi(a, b) < 0$, then $\xi(a, b, c) < 0$ for all $c \in \Gamma$. For any $c \in \mathcal{P}(a)$, the term $-\Delta_{(-\mathbb{R}_+^q)}(\rho(a, c)) \geq 0$, so the minimum in (5.11) equals $\ell(a, b) - \ell(a, c) < 0$, giving $\ell(a, b) < \ell(a, c)$ for all $c \in \mathcal{P}(a)$. This forces $\ell(a, b) < \mathcal{U}(a)$, contradicting $b \in \mathcal{P}(a)$. $\square$

Lemma 5.3 (Characterisation at Lower-Level Optimality [4]). For $a \in \mathcal{D}_a$, $\{(a, b) : b \in \mathcal{P}(a), \ell(a, b) = \mathcal{U}(a)\} = \{(a, b) : b \in \mathcal{P}(a), \Xi(a, b) = 0\}$.

Proof. ($\subseteq$). If $b \in \Lambda(a)$, Lemma 5.2 gives $\Xi(a, b) \geq 0$. If $\Xi(a, b) > 0$, a point $c^* \in \Gamma$ satisfies $\xi(a, b, c^*) > 0$, giving $c^* \in \mathcal{P}(a)$ and $\ell(a, b) > \ell(a, c^*)$, contradicting optimality. So $\Xi(a, b) = 0$.

($\supseteq$). If $\Xi(a, b) = 0$ and $b \in \mathcal{P}(a)$, then $\xi(a, b, c) \leq 0$ for all $c \in \mathcal{P}(a)$. Thus $\ell(a, b) \leq \mathcal{U}(a) \leq \ell(a, b)$, giving equality and $b \in \Lambda(a)$. $\square$

Lemma 5.4 (Active Maximiser Set at a Lower-Level Optimum [4]). If $(\bar{a}, \bar{b})$ is a local weak efficient solution of (MBP), then $Q(\bar{a}) := \{c \in \Gamma : \xi(\bar{a}, \bar{b}, c) = \Xi(\bar{a}, \bar{b})\} = \Lambda(\bar{a})$.

Proof. For any $c \in \Lambda(\bar{a})$: $\ell(\bar{a}, \bar{b}) - \ell(\bar{a}, c) = 0$ and $-\Delta_{(-\mathbb{R}_+^q)}(\rho(\bar{a}, c)) \geq 0$, giving $\xi(\bar{a}, \bar{b}, c) = 0$. No $c^* \in \Gamma$ can have $\xi(\bar{a}, \bar{b}, c^*) > 0$ without providing a feasible point with smaller ℓ -value than $\bar{b}$, contradicting $\bar{b} \in \Lambda(\bar{a})$. Thus $\Xi(\bar{a}, \bar{b}) = 0$ and every element of $\Lambda(\bar{a})$ is an active maximiser. $\square$

5.9.4 The Reformulated Problem

The three lemmas show that $\Xi(a, b) \leq 0$ locally encodes $b \in \Lambda(a)$. The Ξ -reformulated problem is:

$$\begin{aligned} & \mathbb{R}_+^n\text{-min}_{(a,b)} \Phi(a, b) \\ & \text{s.t. } H_j(a, b) \leq 0 \ (j \in \mathcal{M}), \quad \rho_s(a, b) \leq 0 \ (s \in \mathcal{S}), \quad \Xi(a, b) \leq 0. \end{aligned} \tag{RMP}$$

Every locally weak efficient solution of (MBP) is also for (RMP) a locally weak efficient solution, and vice versa [4].

Example 5.5 (Computing Ξ). Let $\ell(a, b) = b^2$, $\rho_1(a, b) = b$, so $\mathcal{P}(a) = \{b \leq 0\}$, $\Lambda(a) = \{0\}$, $\mathcal{U}(a) = 0$, and $\Gamma = [-1, 1]$. For $c \in [-1, 1]$: $\xi(a, b, c) = \min\{b^2 - c^2, \max\{c, 0\}\}$. At $(\bar{a}, \bar{b}) = (a_0, 0)$, one finds $\Xi(a_0, 0) = \max_{c \in [-1, 1]} \xi(a_0, 0, c) = 0$, confirming Lemma 5.3.

5.10 Reduction to an Unconstrained Vector Problem

Set $\mathcal{K} := n + |\mathcal{M}| + |\mathcal{S}| + 1$ and let $\vec{\Phi}(a, b)$ denote the aggregate of $\Phi_i(a, b)$, $H_j(a, b)$, $\rho_s(a, b)$, $\Xi(a, b)$ stacked into a vector in $\mathbb{R}^{\mathcal{K}}$. The *auxiliary unconstrained problem* is

$$\mathbb{R}_+^{\mathcal{K}}\text{-min}_{(a,b)} \vec{\Phi}(a, b). \tag{UMP}$$

Lemma 5.5 (Passage to (UMP) [4]). Every local weak efficient solution of (RMP) is a local weak efficient solution of (UMP) with respect to $\mathbb{R}_+^{\mathcal{K}}$.

Proof. Let $(\bar{a}, \bar{b})$ be locally weak efficient for (RMP) with neighbourhood $\mathcal{N}$. If $(a_k, b_k) \rightarrow (\bar{a}, \bar{b})$ with $\vec{\Phi}(\bar{a}, \bar{b}) - \vec{\Phi}(a_k, b_k) \in \text{int } \mathbb{R}_+^{\mathcal{K}}$, then componentwise: $\Phi(\bar{a}, \bar{b}) - \Phi(a_k, b_k) \in \text{int } \mathbb{R}_+^n$, and all constraints of $(\bar{a}, \bar{b})$ being non-positive forces $(a_k, b_k) \in \mathcal{E} \cap \mathcal{N}$ for large k —contradicting local weak efficiency. $\square$

5.11 The Nonsmooth Abadie Condition and Necessary Optimality Conditions

5.11.1 Linearisation Cone and the NACQ

Denote the constraint components of (RMP) by π_k with index set $\mathcal{P}_0$. At $\bar{w} = (\bar{a}, \bar{b})$, the *active multiplier set* is

$$\mathcal{D}(\bar{w}) := \{t^* \in \mathbb{R}_+^{|\mathcal{P}_0|} : \|t^*\| \leq 1, \sum_k t_k^* \pi_k(\bar{w}) = 0\},$$

and the *linearisation cone* is

$$\mathcal{T}(\bar{w}) := \{d : \forall t^* \in \mathcal{D}(\bar{w}), \forall p_k^* \in \partial^c \pi_k(\bar{w}) : \sum_k t_k^* \langle p_k^*, d \rangle \leq 0\}.$$

Definition 5.8 (Nonsmooth Abadie Constraint Qualification (NACQ) [4]). The NACQ holds at $\bar{w} \in \mathcal{E}$ if $\mathcal{T}(\bar{w}) \subseteq \mathcal{K}(\mathcal{E}, \bar{w})$.

Remark 13. The Ξ -formulation is essential here: unlike the value-function constraint, the Ξ -constraint can satisfy standard qualifications at feasible points. The MFC never holds at feasible points of the value-function reformulation of Ye and Zhu [32], whereas the present reformulation allows it [4].

5.11.2 Necessary Optimality Conditions

Theorem 5.8 (Necessary Conditions: Abstract Form [4]). Let $\bar{w} = (\bar{a}, \bar{b})$ be regarded as locally weak efficient solution of (MBP) at which the NACQ holds. Then $\psi^* \in (-\mathbb{R}_+^n)^\circ \setminus \{0\}$ exists such that

$$0 \in \partial^c(\psi^* \circ \Phi)(\bar{w}) + \overline{\text{co}}\left\{\sum_k t_k^* p_k^* : t^* \in \mathcal{D}(\bar{w}), p_k^* \in \partial^c \pi_k(\bar{w})\right\}. \quad (5.13)$$

Proof. Step 1. $(\bar{a}, \bar{b})$ is qualified as a locally weak efficient solution of (RMP) (via Lemmas 5.2–5.4), so it locally minimises the locally Lipschitz scalar $\tilde{r} := \Delta_{-\text{int } \mathbb{R}_+^n} \circ (\Phi - \Phi(\bar{w}))$ over $\mathcal{E}$. A nonsmooth minimum condition gives $\tilde{r}^* \in \partial^c \tilde{r}(\bar{w})$ with $\langle \tilde{r}^*, d \rangle \geq 0$ for all $d \in \mathcal{K}(\mathcal{E}, \bar{w})$.

Step 2. The NACQ gives $\mathcal{T}(\bar{w}) \subseteq \mathcal{K}(\mathcal{E}, \bar{w})$, so the inequality extends to $\mathcal{T}(\bar{w})$. A polarity argument places $\tilde{r}^*$ in the closed cone in (5.13).

Step 3. The Clarke chain rule applied to $\tilde{r} = \Delta_{-\text{int } \mathbb{R}_+^n} \circ (\Phi - \Phi(\bar{w}))$ yields $\psi^* \in \partial^{ca} \Delta_{-\text{int } \mathbb{R}_+^n}(0)$ with $\tilde{r}^* \in \partial^c(\psi^* \circ \Phi)(\bar{w})$.

Step 4. For $v \in -\mathbb{R}_+^n$: $\langle \psi^*, v \rangle \leq \Delta_{-\text{int } \mathbb{R}_+^n}(v) \leq 0$, so $\psi^* \in (-\mathbb{R}_+^n)^\circ$. Proposition ?? gives $\psi^* \neq 0$. $\square$

Theorem 5.9 (Necessary Conditions: Component Form [4]). Let $(\bar{a}, \bar{b})$ be regarded as locally

weak efficient solution of (MBP). If $\psi^* \in (-\mathbb{R}_+^{\mathcal{K}})^\circ \setminus \{0\}$ exists then

$$0 \in \sum_{i=1}^n \psi_i^* \partial^c \Phi_i(\bar{a}, \bar{b}) + \sum_{j \in \mathcal{M}} \psi_{n+j}^* \partial^c H_j(\bar{a}, \bar{b}) + \sum_{s \in \mathcal{S}} \psi_{n+|\mathcal{M}|+s}^* \partial^c \rho_s(\bar{a}, \bar{b}) + \psi_{\mathcal{K}}^* \partial^c \Xi(\bar{a}, \bar{b}). \quad (5.14)$$

Proof. By Lemma 5.5, $(\bar{a}, \bar{b})$ can be regarded as locally weak efficient solution of (UMP). It locally minimises $\tilde{R} := \Delta_{-\text{int } \mathbb{R}_+^{\mathcal{K}}} \circ (\vec{\Phi} - \vec{\Phi}(\bar{a}, \bar{b}))$ without constraints, giving $0 \in \partial^c \tilde{R}(\bar{a}, \bar{b})$. The Clarke chain rule yields ψ^* as before; $\psi^* \neq 0$ by Proposition ??; the Clarke sum rule applied to $\psi^* \circ \vec{\Phi}$ yields (5.14). $\square$

Remark 14. By Proposition ??, the term $\partial^c \Xi(\bar{a}, \bar{b})$ in (5.14) is expressible through the Clarke subdifferentials of ℓ and ρ_s , making (5.14) fully verifiable from the original data [4].

Example 5.6 (Verification of Theorem 5.9 [4]). Consider the two-objective bilevel problem $\mathbb{R}_+^2$ -min $\Phi(a, b) = (a^2 - b, a + b + 1)$ subject to $H_1(a, b) = -a - 2b \leq 0$, $H_2(a, b) = -b \leq 0$, $b \in \Lambda(a)$, where $\Lambda(a)$ solves $\min_b \ell(a, b) = |a| + b^2$ subject to $\rho_1(a, b) = b \leq 0$.

Analysis. $\mathcal{P}(a) = \{b \leq 0\}$, $\Lambda(a) = \{0\}$, $\mathcal{U}(a) = |a|$; $(\bar{a}, \bar{b}) = (0, 0)$ is feasible with $\Xi(0, 0) = 0$ by Lemma 5.3.

Clarke subdifferentials at $(0, 0)$: $\partial^c \Phi_1 = \{(0, -1)\}$, $\partial^c \Phi_2 = \{(1, 1)\}$, $\partial^c H_1 = \{(-1, -2)\}$, $\partial^c H_2 = \{(0, -1)\}$, $\partial^c \rho_1 = \{(0, 1)\}$, $\partial^c \Xi = \{(0, 0)\}$.

Multiplier check. Take $\psi^* = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2}, 1)$. Then $\frac{1}{2}(0, -1) + \frac{1}{2}(1, 1) + \frac{1}{2}(-1, -2) + \frac{1}{2}(0, -1) + \frac{3}{2}(0, 1) + 1 \cdot (0, 0) = (0, 0)$, confirming (5.14).

5.12 Conclusion

This chapter has produced a unified theory of bilevel optimisation across two levels of generality.

Scalar bilevel (Suneja and Kohli [7]). The equivalence of (BP) and its value-function reformulation (RBP) was established in Lemma 5.1. Five generalised convexity classes (Definitions 5.1–5.3) were introduced and their strict logical independence confirmed by counterexamples. Sufficient optimality conditions were derived under ∂^* -convexity (Theorem 5.2) and under the weaker pair ∂^* -pseudoconvexity/ ∂^* -quasiconvexity (Theorem 5.3). Complete Wolfe-type (Theorems 5.4 and 5.5) and Mond–Weir-type (Theorems 5.6 and 5.7) duality theories were developed. The Mond–Weir dual is applicable under weaker assumptions than the Wolfe dual, reflecting the graduation inherent in the five convexity classes.

Multiobjective bilevel (Gadhi and Dempe [4]). The value-function reformulation fails in this setting because the Mangasarian–Fromovitz condition never holds at feasible points. The Ξ -reformulation (Sections 5.9–5.10) resolves this: three equivalence lemmas (Lemmas 5.2–5.4) show that the constraint $\Xi(a, b) \leq 0$ locally encodes lower-level optimality exactly. The nonsmooth Abadie constraint qualification (Definition 5.8), combined with the non-vanishing

property of the Hiriart-Urruty gauge (Proposition ??) and the Clarke chain rule, yields necessary conditions in abstract form (Theorem 5.8) and in explicit component form (Theorem 5.9).

Connection between the two parts. The scalar bilevel analysis provides the generalised convexity framework that underpins the duality theory of the following chapter, while the Ξ -reformulation of the multiobjective analysis furnishes the reformulated primal problem on which that duality theory is built. Together, the two results form a coherent bridge from single-level convexfactor analysis to the complete duality theorems weak and strong duality, for the bilevel multiobjective programming.

Chapter 6

Duality Results for Bilevel Multiobjective Optimisation via CF

6.1 Introduction

Three lines of analysis developed in earlier chapters come together in this chapter.

From Chapter 3 (Dutta and Chandra [3]). Chapter 3 built the complete theory of convexifactors: Dini derivatives, the four types of convexifactors and their strict hierarchy, the mean-value theorem of Jeyakumar and Luc [19], chain rules for composite functions, the main notions of generalised convexity, and conditions for weak efficiency in single-level vector optimisation. In this chapter the same machinery is brought to bear on bifunctions (a, b) maps to $F_i(a, b)$, $Q_j(a, b)$, $p(a, b)$, $r_s(a, b)$, and $\Psi(a, b)$ at each feasible and dual-feasible point [5, 3].

From Chapter 5 (Suneja and Kohli [7]). & Gadhi and Dempe Chapter 5 applied convexifactors to the scalar bilevel problem. It used the value-function reformulation, introduced five ∂^* -convexity classes for bifunctions, proved necessary and sufficient optimality conditions, and established Wolfe-type and Mond–Weir-type duals with complete weak and strong duality theorems. The present chapter extends that programme to the genuinely multiobjective setting, where the upper-level objective $F = (F_1, \dots, F_n)$ is vector-valued and solutions are characterised by weak efficiency rather than scalar minimality [5, 7, 8].

From Chapter 5 (Gadhi and Dempe [4]). Chapter 5 introduced the Ψ -reformulation, proved its local equivalence to the original bilevel problem via Lemmas, defined the NACQ, and derived necessary conditions using the Clarke subdifferential and the Hiriart-Urruty scalarisation [6]. In this chapter the single-level reformulation (SLP) from Chapter 5 serves as the primal from which the dual (DW) is constructed, and the necessary conditions of Theorem ?? underpin the duality analysis [5].

The principal new result is the combination of the Ψ -reformulation with ∂^* -convexity to obtain the first complete duality theorems for weak and strong the multiobjective bilevel problem in the convexifactor setting. Earlier work by Lafhim et al. [42], Gadhi et al. [43], and Babahadda and Gadhi [28] addressed necessary conditions and partial duality for related models; the present chapter completes that programme [5].

6.2 Preliminaries

All notation follows the unified conventions of Section 2.2 of Chapter 2. The symbols $\mathbb{R}^{n_1}$, $\mathbb{R}^{n_2}$, $\text{cl } A$, $\text{conv } A$, and A° retain their meanings from that section. The definitions collected here are needed throughout; fuller treatments appear in [5, 3, 7, 4, 6, 11, 19].

Remark 15. Throughout this chapter, upper-level constraints are denoted Q_j and lower-level constraints r_s to distinguish them from the scalar bilevel setting of Chapter 4. The lower-level objective $p(a, b)$ and the solution mapping $\Lambda(a)$ retain their notation from Chapter 5.

Preliminary 6.1. Weak Efficient Solution [5, 4, 8]. A point $(\bar{a}, \bar{b}) \in \mathcal{E}$ is a *weak efficient solution* of (BP) if no other feasible (a, b) belongs to E satisfies $F_i(a, b) < F_i(\bar{a}, \bar{b})$ for every i belongs to I simultaneously. When this property holds only within some open neighbourhood of $(\bar{a}, \bar{b})$, the point is a *local weak efficient solution*.

Preliminary 6.2. The Hiriart-Urruty Gauge Δ_C [6, 4, 5]. For a nonempty, closed, convex cone $C \subset \mathbb{R}^n$ with $C \neq \mathbb{R}^n$ and nonempty interior, define $\Delta_C: \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$\Delta_C(a) := \begin{cases} -d(a, \mathbb{R}^n \setminus C) & a \in C, \\ d(a, C) & a \notin C. \end{cases}$$

Proposition 6.1 (Properties of Δ_C) [6, 5]. Let $C \subset \mathbb{R}^n$ be a nonempty, enclosed, $C \neq \mathbb{R}^n$. The function Δ_C is positively homogeneous, convex cone with nonempty interior and 1-Lipschitzian. Moreover, $0 \notin \partial^{ca} \Delta_C(a)$ for every $a \in \mathbb{R}^n$.

Remark 16. The non-vanishing of $\partial^{ca} \Delta_C$ ensures that the Lagrange multiplier for the objective is always nonzero, preventing degeneracy in both the optimality conditions and the dual feasibility conditions [5, 4].

Preliminary 6.3. Dini Directional Derivatives [5, 3, 19]. For F finite at $a \in \mathbb{R}^n$ and direction $u \in \mathbb{R}^n$, the *upper* and *lower Dini directional derivatives* of F at a in direction u are

$$F_d^+(a, u) := \limsup_{t \rightarrow 0^+} \frac{F(a + tu) - F(a)}{t}, \quad F_d^-(a, u) := \liminf_{t \rightarrow 0^+} \frac{F(a + tu) - F(a)}{t}.$$

One always has $F_d^-(a, u) \leq F_d^+(a, u)$, and both are finite when F is locally Lipschitz at a [3].

Preliminary 6.4. Convexifactors of a Bifunction [5, 3, 19]. Let $F: \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \rightarrow \mathbb{R}$ be finite at $(\bar{a}, \bar{b})$.

(i)(i)

1. A closed set $\partial^* F(\bar{a}, \bar{b})$ is an *UCF* of F if

$$F_d^-(\bar{a}, \bar{b}, w) \leq \sup_{\rho^* \in \partial^* F(\bar{a}, \bar{b})} \langle \rho^*, w \rangle \quad \forall w \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}.$$

2. Similarly, it is a *LCF* if

$$F_d^+((\bar{a}, \bar{b}), w) \geq \inf_{\rho^* \in \partial_*^* F(\bar{a}, \bar{b})} \langle \rho^*, w \rangle \quad \forall w.$$

3. The closed set CF of F at $(\bar{a}, \bar{b})$ when it is both a UCF and an LCF.

4. A closed set $\partial^* F(\bar{a}, \bar{b})$ is an *USRCF* when it is a UCF and additionally

$$F_d^+((\bar{a}, \bar{b}), w) \leq \sup_{\rho^* \in \partial^* F(\bar{a}, \bar{b})} \langle \rho^*, w \rangle \quad \forall w.$$

A convexifactor need not be convex or compact, making it applicable to a wider class of nonsmooth functions than the Clarke subdifferential [3, 19]. For any locally Lipschitz F , both $\partial^\circ F(\bar{a}, \bar{b})$ and the Michel–Penot subdifferential [20] are convexifactors of F at $(\bar{a}, \bar{b})$. The *scaling property*: for $\lambda \geq 0$, the set $\lambda \partial^* F(\bar{a}, \bar{b})$ is a convexifactor of λF at $(\bar{a}, \bar{b})$ [34, 7].

Remark 17. The strict hierarchy upper regular $\Rightarrow$ upper semi-regular $\Rightarrow$ upper convexifactor was established in Chapter 3 following [3, 19]. In the dual constraint of Section 6.6, the upper semi-regular convexifactors of F_i appear alongside the upper convexifactors of Q_j , r_s , and Ψ .

Preliminary 6.5. Upper Semicontinuity of Convexifactor Maps [5, 7]. A map $H: \mathbb{R}^n \rightrightarrows \mathbb{R}^m$ is *upper semicontinuous* and being set-valued mapped at $\bar{a}$ if for every $\varepsilon > 0$ $\delta > 0$ such that $H(a) \subseteq H(\bar{a}) + \varepsilon B_{\mathbb{R}^m}$ for all a within distance δ of $\bar{a}$.

Preliminary 6.6. Generalised Convexity via Convexifactors [5, 3, 7]. Let $F: \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \rightarrow \mathbb{R}$ admit a convexifactor $\partial^* F(\bar{a}, \bar{b})$ at $(\bar{a}, \bar{b})$.

(i)(i)

1. **∂^* -Convex.** F is ∂^* -convex at $(\bar{a}, \bar{b})$ if

$$\langle \rho, (a, b) - (\bar{a}, \bar{b}) \rangle \leq F(a, b) - F(\bar{a}, \bar{b}) \quad \forall (a, b), \forall \rho \in \partial^* F(\bar{a}, \bar{b}).$$

The qualifier *strict* applies when strict inequality holds for all $(a, b) \neq (\bar{a}, \bar{b})$.

2. **∂^* -Pseudoconvex.** F is ∂^* -pseudoconvex at $(\bar{a}, \bar{b})$ if

$$F(a, b) < F(\bar{a}, \bar{b}) \implies \langle \rho, (a, b) - (\bar{a}, \bar{b}) \rangle < 0 \quad \forall \rho \in \partial^* F(\bar{a}, \bar{b}).$$

3. **Strictly ∂^* -Pseudoconvex.** F is *strictly* ∂^* -pseudoconvex at $(\bar{a}, \bar{b})$ if for all $(a, b) \neq (\bar{a}, \bar{b})$,

4. **∂^* -Quasiconvex.** F is ∂^* -quasiconvex at $(\bar{a}, \bar{b})$ if

$$F(a, b) \leq F(\bar{a}, \bar{b}) \implies \langle \rho, (a, b) - (\bar{a}, \bar{b}) \rangle \leq 0 \quad \forall \rho \in \partial^* F(\bar{a}, \bar{b}).$$

Remark 18. These four classes extend their classical counterparts to the nonsmooth setting via convexifactors, following [3, 7]. Their mutual relationships are examined in Section 6.3. In this chapter they are applied to each of the data functions of (BP): the objectives F_i , leader's constraints Q_j , the follower's objective p , the follower's constraints r_s , and the function Ψ .

Preliminary 6.7. The Ψ -Reformulation [5, 4, 6]. For fixed a , the feasible set of lower constraint is $Y(a) := \{b : r_s(a, b) \leq 0 \ \forall s \in S\}$ and the value function is $W(a) := \inf\{p(a, b) : b \in Y(a)\}$. With the auxiliary function

$$\psi(a, b, c) := \min\left\{p(a, b) - p(a, c), -\Delta_{(-\mathbb{R}_+^q)}(r_1(a, c), \dots, r_q(a, c))\right\} \quad (6.1)$$

and the Ψ -function $\Psi(a, b) := \max_{c \in Y(a)} \psi(a, b, c)$ (introduced in Chapter 5 following [4]), Lemmas ??-?? of Chapter 5 establish that $\Psi(a, b) \leq 0$ for all feasible (a, b) , with equality precisely when $b \in \Lambda(a)$. The problem (BP) is therefore locally equivalent to the single-level reformulation

$$\begin{aligned} & \mathbb{R}_+^n\text{-Minimize}_{(a,b)} \quad F(a, b) \\ & \text{subject to} \quad Q_j(a, b) \leq 0, \ j \in J, \quad r_s(a, b) \leq 0, \ s \in S, \quad \Psi(a, b) \leq 0. \end{aligned} \quad (\text{SLP})$$

6.3 Interrelations Among the Generalised Convexity Classes

The logical ordering among the four classes of Preliminary 6.7 is recorded here through remarks and examples drawn from [5, 7, 3]. These relationships matter because different duality theorems in Section 6.6 require different subsets of these assumptions.

Remark 19 (∂^* -Convexity Implies ∂^* -Pseudoconvexity [5, 7]). If F is ∂^* -convex and $F(a, b) < F(\bar{a}, \bar{b})$, then ∂^* -convexity gives $\langle \rho, (a, b) - (\bar{a}, \bar{b}) \rangle \leq F(a, b) - F(\bar{a}, \bar{b}) < 0$ for every $\rho \in \partial^* F(\bar{a}, \bar{b})$, establishing ∂^* -pseudoconvexity. The converse fails; the following example shows a ∂^* -pseudoconvex problem that is not ∂^* -convex.

Example 6.1 (∂^* -Pseudoconvex but Not ∂^* -Convex [7]). Define $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$F(a, b) := \begin{cases} \sqrt{a+b}, & a > 0, \ b > 0, \\ \sqrt{a-b}, & a > 0, \ b \leq 0, \\ \sqrt{-a+b}, & a \leq 0, \ b > 0, \\ -\sqrt{-a+b}, & a \leq 0, \ b \leq 0, \end{cases}$$

with convexfactor $\partial^* F(0, 0) = \{(a^*, b^*) : a^* \geq 1, b^* \geq 1\}$. Whenever $F(a, b) < F(0, 0) = 0$ and $\rho \in \partial^* F(0, 0)$, one obtains $\langle \rho, (a, b) \rangle < 0$, so ∂^* -pseudoconvexity holds at $(0, 0)$. Taking $(a, b) = (1, -2)$ and $\rho = (6, 1)$: $F(1, -2) = 1$ while $F(0, 0) + \langle (6, 1), (1, -2) \rangle = 4$, so $F(1, -2) < 4$ gives $\langle \rho, (1, -2) \rangle = 4 > 1 = F(1, -2) - F(0, 0)$, violating the ∂^* -convexity inequality. Hence F is ∂^* -pseudoconvex but not ∂^* -convex at $(0, 0)$ [7].

Remark 20 (∂^* -Pseudoconvexity and ∂^* -Quasiconvexity Are Independent [5, 7]). Neither class implies the other. Examples 6.2 and 6.3 show each direction of non-implication.

Example 6.2 (∂^* -Quasiconvex but Not ∂^* -Pseudoconvex [7]). Define $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$F(a, b) := \begin{cases} \sqrt{a^2 b}, & a \geq 0, b \geq 0, \\ \sqrt{-ab}, & a \geq 0, b < 0, \\ -\sqrt{-ab^2}, & a < 0, b \geq 0, \\ \sqrt{-a^2 b}, & a < 0, b < 0, \end{cases}$$

with convexifactor $\partial^* F(0, 0) = \{(a^*, 0) : a^* \geq 0\}$. At $(-1, 2)$: $F(-1, 2) = 0 = F(0, 0)$ and $\langle (a^*, 0), (-1, 2) \rangle = -a^* \leq 0$ for all $a^* \geq 0$, confirming ∂^* -quasiconvexity. With $\rho = (0, 0) \in \partial^* F(0, 0)$: $F(-1, 2) = 0 \leq F(0, 0)$ but $\langle (0, 0), (-1, 2) \rangle = 0$, which is not strictly negative, so ∂^* -pseudoconvexity fails. Hence F is ∂^* -quasiconvex but not ∂^* -pseudoconvex at $(0, 0)$ [7].

Example 6.3 (∂^* -Pseudoconvex but Not ∂^* -Quasiconvex [7]). Define $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$F(a, b) := \begin{cases} -\sqrt{a} - \sqrt{b}, & a \geq 0, b \geq 0, \\ 0, & a > 0, b < 0, \\ \sqrt{-a} + \sqrt{b}, & a < 0, b \geq 0, \\ \sqrt{-a} + \sqrt{-b}, & a \leq 0, b < 0, \end{cases}$$

with convexifactor $\partial^* F(0, 0) = \{(a^*, b^*) : a^* \leq -1, b^* \leq -1\}$. One verifies ∂^* -pseudoconvexity at $(0, 0)$. At $(1, -1)$ with $\rho = (-1, -2)$: $F(1, -1) = 0 = F(0, 0)$, yet $\langle (-1, -2), (1, -1) \rangle = -1 + 2 = 1 > 0$, violating ∂^* -quasiconvexity. Hence F is ∂^* -pseudoconvex but not ∂^* -quasiconvex at $(0, 0)$ [7].

Remark 21 (Strict ∂^* -Pseudoconvexity Implies ∂^* -Pseudoconvexity [5, 7]). Every strictly ∂^* -pseudoconvex function is ∂^* -pseudoconvex. The converse need not hold. In Example 6.3, $(1, -1) \neq (0, 0)$, $F(1, -1) = 0 \leq F(0, 0)$, but $\langle (-1, -2), (1, -1) \rangle = 1 > 0$, so strict ∂^* -pseudoconvexity fails while ∂^* -pseudoconvexity holds [7].

Example illustrating ∂^* -convexity.

Example 6.4 (∂^* -Convexity for a Quadratic Bifunction). Let $F(a, b) := a^2 + b^2$. At any $(\bar{a}, \bar{b})$, the gradient $\nabla F(\bar{a}, \bar{b}) = (2\bar{a}, 2\bar{b})$ exists, and the singleton $\partial^* F(\bar{a}, \bar{b})$ is a CFr. For all (a, b) :

$$F(a, b) - F(\bar{a}, \bar{b}) = (a - \bar{a})^2 + (b - \bar{b})^2 + 2\bar{a}(a - \bar{a}) + 2\bar{b}(b - \bar{b}) \geq \langle (2\bar{a}, 2\bar{b}), (a, b) - (\bar{a}, \bar{b}) \rangle,$$

with strict inequality for $(a, b) \neq (\bar{a}, \bar{b})$. So F is strictly ∂^* -convex at every point, confirming that smooth convex functions are a special case.

The hierarchy of implications is:

$$\partial^*\text{-convex} \implies \partial^*\text{-pseudoconvex} \implies \partial^*\text{-quasiconvex},$$

with all arrows strict and neither pseudoconvexity nor quasiconvexity implying the other [5, 7].

6.4 Necessary Optimality Conditions

The conditions below combine the Ψ -reformulation of Chapter 5 [4] with the convexfactor calculus of Chapter 3 [3]. Replacing the Clarke subdifferentials used in previous theorem by the (possibly smaller) upper semi-regular convexifactors of F_i and upper convexifactors of Q_j , r_s , and Ψ produces conditions that can be sharper than those of Chapter 5. These conditions also generalise those of Chapter 4: setting $n = 1$ recovers the scalar bilevel case of [7].

Theorem 6.1 (Necessary Optimality Conditions [5, 4]). Suppose that $\bar{u} = (\bar{a}, \bar{b}) \in \mathcal{E}$ be a local weak efficient solution of (BP). Suppose: (i)(i)

1. F_i ($i \in I$) admit bounded upper semi-regular convexifactors $\partial^* F_i(\bar{u})$ at $\bar{u}$ whose set-valued maps are upper semicontinuous at $\bar{u}$;
2. Q_j ($j \in J$) and r_s ($s \in S$) admit upper convexifactors $\partial^* Q_j(\bar{u})$ and $\partial^* r_s(\bar{u})$ at $\bar{u}$;
3. the NACQ holds at $\bar{u}$ wrt $\partial^* Q_j(\bar{u})$ ($j \in J_0$), $\partial^* r_s(\bar{u})$ ($s \in S_0$), and $\partial^* \Psi(\bar{u})$.

Then a multiplier vector $\bar{\pi}^* = (\bar{\lambda}^*, \bar{\mu}^*, \bar{v}^*, \bar{\eta}^*)$ exists with $\bar{\lambda}^* \neq 0_{\mathbb{R}^n}$, satisfying

$$0 \in \sum_{i=1}^n \bar{\lambda}_i^* \partial^* F_i(\bar{u}) + \sum_{j=1}^p \bar{\mu}_j^* \partial^* Q_j(\bar{u}) + \sum_{s=1}^q \bar{v}_s^* \partial^* r_s(\bar{u}) + \bar{\eta}^* \partial^* \Psi(\bar{u}) \quad (6.2)$$

and the complementary slackness conditions

$$\bar{\mu}_k^* Q_k(\bar{u}) = 0 \quad \forall k \in K, \quad \& \bar{v}_s^* r_s(\bar{u}) = 0 \quad \forall s \in S. \quad (6.3)$$

Proof. By Lemmas of Chapter 5, $\bar{u}$ is considered a locally weak efficient solution of the reformulation (SLP). Applying Theorem ?? to (SLP) with the nonsmooth Abadie constraint qualification yields a nonzero vector $\psi^* \in (-\mathbb{R}_+^M)^\circ \setminus \{0\}$ (where $M = n + p + q + 1$) satisfying a zero-inclusion in terms of Clarke subdifferentials. Setting $\bar{\lambda}_i^* := \psi_i^*$, $\bar{\mu}_j^* := \psi_{n+j}^*$, $\bar{v}_s^* := \psi_{n+p+s}^*$, $\bar{\eta}^* := \psi_M^*$ and replacing the Clarke subdifferentials by the upper semi-regular convexifactors of F_i and the upper convexifactors of Q_j , r_s , Ψ via the support-function argument of [3, 19] yields (6.2). The slackness conditions (6.3) follow from the inner-product argument in Theorem ?. $\square$

Remark 22. When $n = 1$, Theorem 6.1 reduces to the necessary conditions of Chapter 4 [7]. When convexifactors coincide with Clarke subdifferentials, it gives exactly Theorem ?. The sharper result arises because $\text{co } \partial^* F_i(\bar{u})$ may be strictly contained in $\partial^\circ F_i(\bar{u})$ [3, 19].

6.5 Sufficient Optimality Conditions

Theorem 6.2. [5, 43] Suppose that $\bar{u} = (\bar{a}, \bar{b}) \in \mathcal{E}$ be feasible for (BP). Suppose a multiplier vector $\bar{\pi}^* = (\bar{\lambda}^*, \bar{\mu}^*, \bar{v}^*, \bar{\eta}^*)$ with $\bar{\lambda}^* \neq 0$ satisfies (6.2) and (6.3), and that F_i ($i \in I$), Q_j ($j \in J_0(\bar{u})$), r_s ($s \in S_0(\bar{u})$), and Ψ are ∂^* -convex at $\bar{u}$. Then $(\bar{a}, \bar{b})$ is weakly efficient of (BP).

Proof. Let that since $\bar{\lambda}^* \geq 0$ and $\bar{\lambda}^* \neq 0$:

$$\sum_{i=1}^n \bar{\lambda}_i^* F_i(a, b) < \sum_{i=1}^n \bar{\lambda}_i^* F_i(\bar{u}). \quad (6.4)$$

for contradiction. From (6.2), elements $\xi_i \in \partial^* F_i(\bar{u})$, $\eta_j \in \partial^* Q_j(\bar{u})$, $\theta_s \in \partial^* r_s(\bar{u})$, $\chi \in \partial^* \Psi(\bar{u})$ exist with

$$0 = \sum_{i=1}^n \bar{\lambda}_i^* \xi_i + \sum_{j=1}^p \bar{\mu}_j^* \eta_j + \sum_{s=1}^q \bar{v}_s^* \theta_s + \bar{\eta}^* \chi. \quad (6.5)$$

Step 1. ∂^* -convexity of F_i at $\bar{u}$: $\langle \xi_i, (a, b) - \bar{u} \rangle \leq F_i(a, b) - F_i(\bar{u})$. Multiplying by $\bar{\lambda}_i^* \geq 0$ and summing over i :

$$\left\langle \sum_{i=1}^n \bar{\lambda}_i^* \xi_i, (a, b) - \bar{u} \right\rangle \leq \sum_{i=1}^n \bar{\lambda}_i^* (F_i(a, b) - F_i(\bar{u})) < 0, \quad (6.6)$$

using (6.4) in the final step.

Step 2. For $j \in J_0(\bar{u})$: ∂^* -convexity of Q_j , primal feasibility $Q_j(a, b) \leq 0$, and $Q_j(\bar{u}) = 0$. For $j \notin J_0(\bar{u})$: $\bar{\mu}_j^* = 0$ by (6.3). So $\langle \sum_j \bar{\mu}_j^* \eta_j, (a, b) - \bar{u} \rangle \leq 0$. The same argument for r_s using $S_0(\bar{u})$ and $\bar{v}_s^*$ gives $\langle \sum_s \bar{v}_s^* \theta_s, (a, b) - \bar{u} \rangle \leq 0$.

Step 3. By Lemma, $(a, b) \in \mathcal{E}$ implies $\Psi(a, b) \leq 0$; by Lemma, $\bar{b} \in \Lambda(\bar{a})$ implies $\Psi(\bar{u}) = 0$. ∂^* -convexity of Ψ : $\langle \bar{\eta}^* \chi, (a, b) - \bar{u} \rangle \leq \bar{\eta}^* (\Psi(a, b) - \Psi(\bar{u})) \leq 0$.

Step 4. Taking the inner product of (6.5) with $(a, b) - \bar{u}$ and applying Steps 1–3:

$$0 = \left\langle \sum_i \bar{\lambda}_i^* \xi_i + \sum_j \bar{\mu}_j^* \eta_j + \sum_s \bar{v}_s^* \theta_s + \bar{\eta}^* \chi, (a, b) - \bar{u} \right\rangle < 0,$$

a contradiction. Hence $(\bar{a}, \bar{b})$ is a weak efficient solution. $\square$

Example illustrating sufficient conditions.

Example 6.5 (Verification of Sufficient Optimality Conditions). Consider (BP) with $F(a, b) = (a^2 - b, a + b)$, $Q_1(a, b) = a - 1 \leq 0$, and lower-level $\min_b p(a, b) = b^2$ subject to $r_1(a, b) = b \leq 0$, giving $\Lambda(a) = \{0\}$ for all a . The feasible set is $\mathcal{E} = \{(a, 0) : a \leq 1\}$ with $\Psi(a, 0) = 0$.

At $\bar{u} = (0, 0)$: $Q_1(0, 0) = -1 < 0$ (inactive, $\bar{\mu}_1^* = 0$), $r_1(0, 0) = 0$ (active), $\Psi(0, 0) = 0$. Convexifactors: $\partial^* F_1(0, 0) = \{(0, -1)\}$, $\partial^* F_2(0, 0) = \{(1, 1)\}$, $\partial^* r_1(0, 0) = \{(0, 1)\}$,

$\partial^*\Psi(0,0) = \{(0,0)\}$. With $\bar{\lambda}_1^* = 1, \bar{\lambda}_2^* = 0, \bar{\nu}_1^* = 1, \bar{\eta}^* = 0$: $1 \cdot (0, -1) + 1 \cdot (0, 1) = (0, 0)$, confirming (6.2). The functions F_1, r_1 , and Ψ are ∂^* -convex at $(0,0)$. By Theorem 6.2, $(0,0)$ is a weak efficient solution. On $\mathcal{E}$: $F(a,0) = (a^2, a)$; reducing a lowers F_1 but raises $-F_2$, so no feasible point dominates $(0,0)$ in both components.

6.6 Duality Results

6.6.1 Formulation of the Dual Problem of Wolfe-Type

A Wolfe-type dual for the bilevel primal (BP) (equivalently (SLP)) is constructed following Saini and Kailey [5]. The dual variable is $(c, d) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ and the multiplier vector is $\pi^* = (\lambda_1^*, \dots, \lambda_n^*, \mu_1^*, \dots, \mu_p^*, \nu_1^*, \dots, \nu_q^*, \eta^*)$.

Definition 6.1 (Wolfe-Type Dual Problem (DW) [5]). The Wolfe-type dual of (BP) is:

$$\begin{aligned} & \mathbb{R}_+^n\text{-Maximize}_{(c,d), \pi^*} \quad \theta((c, d), \pi^*) \\ & \text{subject to} \quad 0 \in \sum_{i=1}^n \lambda_i^* \partial^* F_i(c, d) + \sum_{j=1}^p \mu_j^* \partial^* Q_j(c, d) + \sum_{s=1}^q \nu_s^* \partial^* r_s(c, d) + \eta^* \partial^* \Psi(c, d), \\ & \quad \pi^* \geq 0, \quad (\lambda_1^*, \dots, \lambda_n^*) \neq (0, \dots, 0), \quad \sum_{i=1}^n \lambda_i^* = 1, \end{aligned} \tag{DW}$$

where the i -th component of the dual objective is

$$\theta_i((c, d), \pi^*) := F_i(c, d) + \sum_{j=1}^p \mu_j^* Q_j(c, d) + \sum_{s=1}^q \nu_s^* r_s(c, d) + \eta^* \Psi(c, d), \quad i \in I. \tag{6.7}$$

The feasible set of (DW) is denoted $\widehat{E}$.

Remark 23 (Non-emptiness of $\widehat{E}$ [5]). The set $\widehat{E}$ is nonempty. Theorem 6.1 produces multipliers $\bar{\pi}^*$ satisfying the zero-inclusion (6.2) at $\bar{u}$ with $\bar{\lambda}^* \neq 0$. After normalising $\bar{\lambda}^*$ maps to $\bar{\lambda}^*/\sum_i \bar{\lambda}_i^*$, the condition $\sum_i \bar{\lambda}_i^* = 1$ is satisfied and $(\bar{u}, \bar{\pi}^*) \in \widehat{E}$.

Example illustrating the dual structure.

Example 6.6 (Structure of the Wolfe Dual). Continuing Example 6.5 with $F_1(a, b) = a^2 - b$, $F_2(a, b) = a + b$, $Q_1(a, b) = a - 1$, $r_1(a, b) = b$. At $(c, d) = (0, 0)$ with $\lambda^* = (1, 0)$, $\nu_1^* = 1, \mu_1^* = \eta^* = 0$: the stationarity condition gives $0 \in 1 \cdot (0, -1) + 0 \cdot (1, 1) + 1 \cdot (0, 1) = (0, 0)$, confirming $(c, d, \pi^*) = ((0, 0), (1, 0, 0, 1, 0)) \in \widehat{E}$. The dual objective at this point is $\theta_1((0, 0), \pi^*) = F_1(0, 0) = 0 = F_1(\bar{u})$, anticipating the objective equality to be established in strong duality.

6.6.2 Weak Duality

Theorem 6.3 (Weak Duality Theorem [5]). Let $(a, b) \in \mathcal{E}$ be feasible for (BP) and let $((c, d), \pi^*) \in \widehat{E}$ be feasible for (DW). If F_i ($i \in I$), Q_j ($j \in J$), r_s ($s \in S$), and Ψ are all ∂^* -convex at (c, d) , then $F(a, b) \not\leq \theta((c, d), \pi^*)$.

Proof. Suppose for contradiction that $F_i(a, b) \leq \theta_i((m, n), \pi^*)$ for all $i \in I$. Since $\lambda^* \geq 0$, $\lambda^* \neq 0$, and $\sum_i \lambda_i^* = 1$:

$$\sum_{i=1}^n \lambda_i^* F_i(a, b) \leq \sum_{i=1}^n \lambda_i^* \theta_i((m, n), \pi^*). \quad (6.8)$$

From dual feasibility, elements $\xi_i \in \partial^* F_i(m, n)$, $\eta_j \in \partial^* Q_j(m, n)$, $\theta_s \in \partial^* r_s(m, n)$, $\chi \in \partial^* \Psi(m, n)$ exist satisfying

$$0 = \sum_{i=1}^n \lambda_i^* \xi_i + \sum_{j=1}^p \mu_j^* \eta_j + \sum_{s=1}^q \nu_s^* \theta_s + \eta^* \chi. \quad (6.9)$$

Step 1. ∂^* -convexity of F_i at (c, d)

Step 2. ∂^* -convexity of each Q_j at (c, d) and primal feasibility $Q_j(a, b) \leq 0$ give $\langle \eta_j, (a, b) - (c, d) \rangle \leq Q_j(a, b) - Q_j(c, d) \leq -Q_j(c, d)$. Weighting by $\mu_j^* \geq 0$:

$$\left\langle \sum_{j=1}^p \mu_j^* \eta_j, (a, b) - (c, d) \right\rangle \leq - \sum_{j=1}^p \mu_j^* Q_j(c, d).$$

Step 3. Substitute (6.9) into (??) and apply Step 2:

$$\begin{aligned} \sum_{i=1}^n \lambda_i^* F_i(a, b) &\geq \sum_{i=1}^n \lambda_i^* F_i(m, n) - \left\langle \sum_j \mu_j^* \eta_j + \sum_s \nu_s^* \theta_s + \eta^* \chi, (a, b) - (m, n) \right\rangle \\ &\geq \sum_{i=1}^n \lambda_i^* F_i(m, n) + \sum_{j=1}^p \mu_j^* Q_j(m, n) + \sum_{s=1}^q \nu_s^* r_s(m, n) + \eta^* \Psi(m, n) \\ &= \sum_{i=1}^n \lambda_i^* \theta_i((m, n), \pi^*), \end{aligned} \quad (6.10)$$

using definition (6.7) and $\sum_i \lambda_i^* = 1$ in the last step. This contradicts (6.8). $\square$

Remark 24 (Interpretation of Weak Duality [5, 8]). Theorem 6.3 is the multiobjective analogue of the classical weak duality bound: no primal feasible point can component-wise dominate any dual feasible point under the cone order of $\mathbb{R}_+^n$. The dual problem (DW) therefore provides a meaningful bound on the primal (BP) [11].

Example illustrating weak duality.

Example 6.7 (Weak Duality). Continuing Examples 6.5 and 6.6: the primal feasible set is $\mathcal{E} = \{(a, 0) : a \leq 1\}$ with $F(a, 0) = (a^2, a)$, and the dual feasible point is $((c, d), \pi^*) = ((0, 0), (1, 0, 0, 1, 0))$ with $\theta((0, 0), \pi^*) = (0, 0)$. For any primal $(a, 0)$ with $a > 0$: $F_1(a, 0) = a^2 > 0 = \theta_1$, so $F(a, 0) \not\leq \theta((0, 0), \pi^*)$. For $a < 0$: $F_1(a, 0) = a^2 > 0$, again preventing domination. At $a = 0$ the objectives coincide, which is consistent with the equality in strong duality below.

6.6.3 Strong Duality

Theorem 6.4 (Strong Duality Theorem [5]). Suppose that $\bar{u} = (\bar{a}, \bar{b}) \in \mathcal{E}$ be considered weakly efficient at (BP) at which the NACQ holds. Then a multiplier vector $\bar{\pi}^*$ exists such that $(\bar{u}, \bar{\pi}^*) \in \widehat{E}$ and $F(\bar{a}, \bar{b}) = \theta(\bar{u}, \bar{\pi}^*)$. Moreover, if F_i ($i \in I$), Q_j ($j \in J$), r_s ($s \in S$), and Ψ are ∂^* -convex at $\bar{u}$, then $(\bar{u}, \bar{\pi}^*)$ is a weak efficient solution of (DW).

Proof. Dual feasibility. Theorem 6.1 furnishes $\bar{\pi}^*$ satisfying (6.2) with $\bar{\lambda}^* \neq 0$. Normalising $\bar{\lambda}^*$ maps to $\bar{\lambda}^*/\kappa$ where $\kappa := \sum_i \bar{\lambda}_i^* > 0$, and scaling $\bar{\mu}^*, \bar{v}^*, \bar{\eta}^*$ by $1/\kappa$ (which preserves the zero-inclusion by linearity), achieves $\sum_i \bar{\lambda}_i^* = 1$. Hence $(\bar{u}, \bar{\pi}^*) \in \widehat{E}$.

Objective equality. Since $\bar{b} \in \Lambda(\bar{a})$, Lemma ?? gives $\Psi(\bar{u}) = 0$. Complementary slackness (6.3) gives $\bar{\mu}_j^* Q_j(\bar{u}) = 0$ for all j and $\bar{v}_s^* r_s(\bar{u}) = 0$ for all s . By definition (6.7): $\theta_i(\bar{u}, \bar{\pi}^*) = F_i(\bar{u})$ for each $i \in I$.

Weak efficiency of the dual point. Suppose $(\bar{u}, \bar{\pi}^*)$ is not weakly efficient for (DW). Then some $((c, d), \pi^*) \in \widehat{E}$ satisfies $\theta_i((c, d), \pi^*) > F_i(\bar{u})$ for all $i \in I$. Applying Theorem 6.3 with primal point $\bar{u}$ and dual point $((c, d), \pi^*)$ gives $F(\bar{u}) \not\leq \theta((c, d), \pi^*)$, contradicting the strict dominance assumption. $\square$

Remark 25 (Significance of Strong Duality [5, 7]). Theorem 6.4 is the principal theoretical result of this chapter. Three steps are essential: the necessary conditions of Theorem 6.1 establish dual feasibility; the identities $\Psi(\bar{u}) = 0$ and the slackness conditions (6.3) force the primal and dual objective values to coincide; and Theorem 6.3 rules out the existence of a strictly better dual point. The convexfactor language executes all three steps, with the possible sharpening over the Clarke subdifferential following from the fact that $\text{co } \partial^* F_i(\bar{u})$ may be strictly contained in $\partial^\circ F_i(\bar{u})$ [5, 3].

6.7 Illustrative Examples for the Duality Results

6.7.1 Illustrative Example from the Literature

Example 6.8 (The Bilevel Multiobjective Problem and Its Wolfe Dual [5]). Consider

$$\begin{aligned} \mathbb{R}_+^2 \text{-} \min_{(a,b)} \quad & P(a, b) = (a^2 - b, a + b + 1) \\ \text{s.t.} \quad & Q_1(a, b) = -a - 2b \leq 0, \quad Q_2(a, b) = -b \leq 0, \quad b \in \Lambda(a), \end{aligned} \quad (\text{Q})$$

where $\Lambda(a)$ solves $\min_b p(a, b) = |a| + b^2$ subject to $r_1(a, b) = b \leq 0$.

Lower-level analysis. $Y(a) = \{b \leq 0\}$, $\Lambda(a) = \{0\}$ for all $a \in \mathbb{R}$, and $W(a) = |a|$.

Feasibility of $\bar{u} = (0, 0)$. $Q_1(0, 0) = Q_2(0, 0) = 0$ (both active), $r_1(0, 0) = 0$ (active), $0 \in \Lambda(0)$, and $\Psi(0, 0) = 0$ by Lemma ???. So $\bar{u}$ is feasible for (SLP). The feasible set is $\mathcal{E} = \mathbb{R}_+ \times \{0\}$.

Weak efficiency. On $\mathcal{E}$: $F(a, 0) = (a^2, a + 1)$. For $a > 0$: F_1 increases. For $a < 0$: F_2 decreases. No feasible point strictly lowers both components simultaneously, so $\bar{u}$ is weakly efficient.

Active sets. $J_0(\bar{u}) = \{1, 2\}$, $S_0(\bar{u}) = \{1\}$.

Convexifactors at $\bar{u} = (0, 0)$.

$$\begin{aligned} \partial^* F_1(0, 0) &= \{(0, -1)\}, & \partial^* F_2(0, 0) &= \{(1, 1)\}, \\ \partial^* Q_1(0, 0) &= \{(-1, -2)\}, & \partial^* Q_2(0, 0) &= \{(0, -1)\}, \\ \partial^* r_1(0, 0) &= \{(0, 1)\}, & \partial^* \Psi(0, 0) &= \{(0, 0)\}. \end{aligned}$$

All six functions are ∂^* -convex at $\bar{u}$ [5].

NACQ verification. $C(\bar{u}) = \{(-1, -2), (0, -1), (0, 1), (0, 0)\}$ gives $[C(\bar{u})]^\circ = \mathbb{R}_+ \times \{0\} = K(\mathcal{E}, \bar{u})$, so the NACQ holds.

Multiplier verification. Take $\bar{\pi}^* = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2}, 1)$, i.e. $\bar{\lambda}_1^* = \bar{\lambda}_2^* = \frac{1}{2}$, $\bar{\mu}_1^* = \bar{\mu}_2^* = \frac{1}{2}$, $\bar{\nu}_1^* = \frac{3}{2}$, $\bar{\eta}^* = 1$, with $\sum_i \bar{\lambda}_i^* = 1$:

$$\frac{1}{2}(0, -1) + \frac{1}{2}(1, 1) + \frac{1}{2}(-1, -2) + \frac{1}{2}(0, -1) + \frac{3}{2}(0, 1) + 1 \cdot (0, 0) = (0, 0). \quad \checkmark$$

The Wolfe dual (DW) for (Q).

$$\mathbb{R}_+^2 \text{-} \max_{(c,d), \pi^*} (\theta_1(c, d, \pi^*), \theta_2(c, d, \pi^*))$$

$$\begin{aligned} \text{s.t.} \quad & 0 \in \lambda_1^* \partial^* F_1(c, d) + \lambda_2^* \partial^* F_2(c, d) + \mu_1^* \partial^* Q_1(c, d) + \mu_2^* \partial^* Q_2(c, d) + \nu_1^* \partial^* r_1(c, d) + \eta^* \partial^* \Psi(c, d), \\ & \pi^* \geq 0, \quad (\lambda_1^*, \lambda_2^*) \neq (0, 0), \quad \lambda_1^* + \lambda_2^* = 1. \end{aligned}$$

(DW_Q)

A second dual-feasible point at $(c, d) = (-1, 0)$. Convexifactors at $(-1, 0)$: $\partial^* F_1(-1, 0) = \{(-2, -1)\}$, $\partial^* F_2(-1, 0) = \{(1, 1)\}$, $\partial^* Q_1(-1, 0) = \{(-1, -2)\}$, $\partial^* Q_2(-1, 0) = \{(0, -1)\}$,

$\partial^* r_1(-1, 0) = \{(0, 1)\}$, $\partial^* \Psi(-1, 0) = \{(0, 0)\}$. With $\pi^* = (\frac{1}{4}, \frac{3}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 1)$:

$$\frac{1}{4}(-2, -1) + \frac{3}{4}(1, 1) + \frac{1}{4}(-1, -2) + \frac{1}{2}(0, -1) + \frac{1}{4}(0, 1) + 1 \cdot (0, 0) = (0, 0). \checkmark$$

So $((-1, 0), \pi^*) \in \widehat{E}$ and all data functions are ∂^* -convex at $(-1, 0)$ [5].

Weak duality. For any primal $(a, 0) \in \mathcal{E}$ and dual $((-1, 0), \pi^*)$, Theorem 6.3 gives $F(a, 0) \not\leq \theta(-1, 0, \pi^*)$.

Strong duality. By Theorem 6.4, $(\bar{u}, \bar{\pi}^*) = ((0, 0), \bar{\pi}^*)$ is feasible for (DW) with $F(\bar{u}) = \theta(\bar{u}, \bar{\pi}^*)$, and the weak efficient solution being $(\bar{u}, \bar{\pi}^*)$ of (DW) [5].

6.7.2 Illustrative Example: Wolfe Dual for a Two-Objective Bilevel Problem

Example 6.9 (Wolfe Dual: Two-Objective Bilevel Problem). Consider

$$\begin{aligned} \mathbb{R}_+^2\text{-min}_{(a,b)} F(a, b) &= (a^2 + b, -a + b^2) \\ \text{s.t. } Q_1(a, b) &= a - 1 \leq 0, \quad b \in \Lambda(a), \end{aligned}$$

where $\Lambda(a)$ solves $\min_b p(a, b) = b^2$ subject to $r_1(a, b) = b \leq 0$.

Lower-level analysis. $Y(a) = \{b \leq 0\}$, $\Lambda(a) = \{0\}$ for all a , $W(a) = 0$.

Feasibility and objective at $\bar{u} = (0, 0)$. $Q_1(0, 0) = -1 < 0$ (inactive), $r_1(0, 0) = 0$ (active), $\Psi(0, 0) = 0$. On $\mathcal{E} = \{(a, 0) : a \leq 1\}$: $F(a, 0) = (a^2, -a)$.

Weak efficiency. For $a > 0$: $F_2(a, 0) = -a < 0 = F_2(0, 0)$, so F_1 is not improved. Hence $(0, 0)$ is weakly efficient.

Convexifactors at $(0, 0)$. $\partial^* F_1(0, 0) = \{(0, 1)\}$, $\partial^* F_2(0, 0) = \{(-1, 0)\}$, $\partial^* r_1(0, 0) = \{(0, 1)\}$, $\partial^* \Psi(0, 0) = \{(0, 0)\}$, $\partial^* Q_1$ is irrelevant since Q_1 is inactive.

Multipliers. Adjusting: $\bar{\lambda}_1^* = 0$, $\bar{\lambda}_2^* = 1$, $\bar{v}_1^* = 0$, gives $0 \in (-1, 0)$, which fails. A valid selection uses $\bar{\lambda}_1^* = \frac{1}{2}$, $\bar{\lambda}_2^* = \frac{1}{2}$, $\bar{v}_1^* = \frac{1}{2}$: $\frac{1}{2}(0, 1) + \frac{1}{2}(-1, 0) + \frac{1}{2}(0, 1) = (\frac{1}{2}, 1) \neq (0, 0)$. The correct multipliers satisfying (6.2) are $\bar{\lambda}_1^* = 1$, $\bar{\lambda}_2^* = 0$, $\bar{v}_1^* = 1$: $1 \cdot (0, 1) + 1 \cdot (0, -1) = (0, 0)$. $\checkmark$ (Here r_1 contributes $(0, -1)$? No: $\partial^* r_1(0, 0) = \{(0, 1)\}$. On Checking: $\bar{\lambda}_2^*$ must be nonzero: $\bar{\lambda}_1^*(0, 1) + \bar{\lambda}_2^*(-1, 0) + \bar{v}_1^*(0, 1) = (\bar{\lambda}_2^*, \bar{\lambda}_1^* + \bar{v}_1^*) = (0, 0)$. This gives $\bar{\lambda}_2^* = 0$ and $\bar{\lambda}_1^* + \bar{v}_1^* = 0$, which is contradicting)

This example illustrates an important structural point: the Ψ -function at $\bar{u}$ has $\partial^* \Psi(0, 0) = \{(0, 0)\}$ (trivial subdifferential), so $\bar{\eta}^* \chi = (0, 0)$ contributes nothing. The burden of satisfying (6.2) falls entirely on the objective and constraint convexifactors. When no nonnegative multipliers resolve the zero-inclusion, the feasible point fails the NACQ (its contingent cone is too small), and the necessary conditions of Theorem 6.1 do not apply. This confirms that the NACQ is a genuine restriction, not automatically satisfied.

For the primal-dual objective match to hold, the data must be arranged so that both the

NACQ and a nontrivial KKT system are satisfied simultaneously. Example 6.8 from [5] is the canonical illustration of this.

6.8 Concluding Remarks

This chapter has provided a comprehensive, stand-alone, and rigorous duality for the bilevel multiobjective problem (??) optimisation problem (??) in the convexifactor approach, as in Saini and Kailey [5].

Building on previous chapters. The chapter builds on all three of the previous technical chapters: the calculus of Chapter 3 [3, 19], the generalized classes and bilevel duality of Chapter 4 [7], and the Ψ -reformulation and necessary optimality conditions of Chapter 5 [4]. The present chapter is the last chapter of the doctoral thesis' theoretical journey by deals with duality in the truly multiobjective bilevel problem.

Paper's content fully covered. All definitions (Wolfe dual, ∂^* -convexity classes), all theorems (necessary, sufficient, weak duality, strong duality), all propositions (Δ_C properties), all examples of the original paper (Example 6.8), and all interrelation remarks with counterexamples (Remarks 19–21, Examples 6.1–6.3) are included.

Generalized convexity interrelations. The interrelation ∂^* -convex $\Rightarrow$ strictly ∂^* -pseudoconvex $\Rightarrow \partial^*$ -pseudoconvex, as well as the independence of ∂^* -pseudoconvexity and ∂^* -quasiconvexity, was shown with four remarks and three counterexamples, all taken from [7]. These interrelations explain why ∂^* -convexity is the right assumption for the Wolfe dual, but a weaker condition is sufficient for the (scalar) Mond–Weir dual in Chapter 4.

Duality theorems. Weak duality (Theorem 6.3) proves the basic cone ordering of the primal and dual objectives with the ∂^* -convexity of all data functions at the dual point. Strong duality (Theorem 6.4) implies that every primal weak solution with the NACQ extends to a dual weak efficient solution solution with the same objective values - the full strong duality result in duality result in this nonsmooth multiobjective bilevel framework.

Future directions. As Saini and Kailey [5] noted, the current framework can be applied to nonsmooth bilevel programming problems with involve vector-valued objectives. The approach used in this dissertation (Ψ -reformulation, convexifactor calculus, convexity, convexity, duality—provides all the necessary tools for such an extension, and opens further questions in sensitivity analysis, stability analysis, and algorithmic design for nonsmooth hierarchical multiobjective optimisation.

Chapter 7

Comparative Analysis and Conclusion

7.1 Comparative Analysis

7.1.1 Comparison Based on Problem Type and Nature of Results

This table compares the type and level of optimisation problem considered in each paper and the nature of the results obtained.

Paper	Type of Problem	Level	Nature of Results
Dutta & Chandra (2004) [3]	Constrained vector minimisation with locally Lipschitz objectives and constraints	Single-level; multi-objective	Necessary optimality conditions for weak efficient points
Suneja & Kohli (2011) [7]	Bilevel programming with scalar upper-level and lower-level objectives; locally Lipschitz data	Bilevel; single-objective	Necessary and sufficient optimal conditions; Wolfe-type and Mond-Weir type duality (weak and strong)
Gadhi & Dempe (2012) [4]	Bilevel programming with vector-valued scalar lower-level and upper level objective; locally Lipschitz data	Bilevel; multiobjective	Necessary optimal conditions for local weak efficient solutions only
Saini & Kailey (2024) [5]	Bilevel programming with vector-valued scalar lower-level and upper level objective; locally Lipschitz data	Bilevel; multiobjective	Necessary and sufficient optimality conditions; Wolfe-type duality (weak and strong)

Table 7.1: Comparison of the four papers based on problem type and nature of results.

7.1.2 Comparison Based on Convexifactors and Mathematical Techniques

This table compares how convexifactors are used in each paper, along with the primary mathematical techniques employed and the key contribution of each work.

Paper	Role of Convexifactors	Techniques Used	Key Contribution
Dutta & Chandra (2004) [3]	Central: full calculus developed here; used in necessary conditions via support-function and separation arguments	Upper/lower Dini derivatives; mean-value theorem; chain rules (bounded and unbounded cases); ∂ -pseudoconvexity; pseudomonotonicity	First systematic convexifactor calculus; sharper necessary conditions for vector optimisation than Clarke subdifferential
Suneja & Kohli (2011) [7]	Central: five ∂^* -convexity classes defined; convexifactors appear in optimality conditions and in both dual problem constraints	Value function reformulation ($\phi - V \leq 0$); ∂^* -convex, ∂^* -pseudoconvex, ∂^* -quasiconvex bifunctions; Wolfe and Mond–Weir dual objectives; complementary slackness	First bilevel duality via convexifactors; complete Wolfe and Mond–Weir framework for scalar bilevel programming
Gadhi & Dempe (2012) [4]	Absent: Clarke subdifferential used throughout; convexifactors not employed	Ψ -reformulation; Hiriart-Urruty scalarisation Δ_S ; Clarke calculus (max, sup, chain rules); nonsmooth Abadie CQ; local equivalence lemmas	Ψ -reformulation for multiobjective bilevel problems; Mangasarian–Fromovitz CQ can hold at feasible points; KKT conditions without partial calmness
Saini & Kailey (2024) [5]	Central: upper semi-regular convexifactors of objectives; upper convexifactors of constraints; combined with Ψ -reformulation for duality	Ψ -reformulation (from [4]); convexifactor calculus (from [3]); ∂^* -convexity (from [7]); nonsmooth Abadie CQ (convexifactor version)	First multiobjective bilevel duality via convexifactors; unification of all three preceding frameworks

Table 7.2: Comparison of the four papers based on convexifactors and mathematical techniques.

7.2 Discussion of Comparative Analysis

A combination of the two tables indicates four major trends in the evolution of the field.

Developing from simple to complex problems. The issues discussed in the four articles become progressively more general in a systematic and consciously planned manner, as is

evident in Table 1. Dutta and Chandra [3] work at the most elementary level: a single-level minimisation problem of a vector with an explicit feasible set and no hierarchical structure. Suneja and Kohli [7] first introduce the bilevel structure, but limit the upper-level goal to a scalar function, so that optimality is still given in a classical form. Gadhi and Dempe [4] restore the upper-level objective to a vector, and the problem becomes simultaneously bilevel and multiobjective. Lastly, Saini and Kailey [5] integrate all three characteristics—a bilevel structure, multiple objectives, and convexifactor-based analysis—into one framework. This stepwise progression reflects the actual mathematical challenge of the fully general problem: with each additional feature, new analytical issues arise at the lower level before the next stage can be addressed.

Evolution of the role of convexifactors. Table 2 indicates that the role of convexifactors changes qualitatively from paper to paper. In [3], they are the main object of study, and the entire paper is devoted to constructing their calculus. In [7], the calculus is taken as given and used to specify generalized classes of convexity and to build dual problems; convexifactors are no longer the object of study but a tool. In [4], they are explicitly omitted: the Clarke subdifferential is used instead, and the introduction of the Ψ -reformulation makes the analysis technically cleaner. In [5], convexifactors reappear and are combined with the Ψ -reformulation for the first time, signalling that the reformulation framework has become mature enough to support the added generality of convexifactors. Their reappearance in the most general paper is important because it strengthens the analysis beyond the Clarke framework in the multiobjective bilevel setting.

Difference in techniques applied. The most important technical contrast visible in Table 2 is the difference between the value-function reformulation used in [7] and the Ψ -reformulation introduced in [4]. The value-function constraint $\phi(a, b) - V(a) \leq 0$ is active at all feasible points, causing the Mangasarian–Fromovitz constraint qualification to fail, and it therefore requires a separate regularity condition to ensure that the objective multiplier is nonzero. The Ψ -function, constructed using the Hiriart-Urruty scalarisation Δ_S , encodes the same lower-level optimality condition as a locally Lipschitz scalar inequality, for which the usual constraint qualifications can hold. This allows the KKT framework to be applied more cleanly and without additional complications. Saini and Kailey [5] adopt the Ψ -reformulation precisely to gain this constraint-qualification advantage, and it is this reformulation, combined with the convexifactor framework, that makes the complete duality theory possible.

From optimality to duality. Table 1 shows that two of the four papers, namely Dutta and Chandra [3] and Gadhi and Dempe [4], concentrate mainly on optimality conditions, whereas Suneja and Kohli [7] and Saini and Kailey [5] develop full duality theories. Duality requires both a sound reformulation that produces the dual constraints and an appropriate convexity condition on the data in order to establish weak and strong duality. The reformulation is provided by Gadhi and Dempe in the multiobjective context, while Dutta and Chandra supply the convexity framework in the single-level setting. Saini and Kailey are the first to unite

these two ingredients in the multiobjective bilevel setting, which is why the duality results of Chapter 6 form the culminating theoretical contribution of this dissertation.

7.3 Evolution of the Topic

The overall development of the subject can be captured in the following conceptual flow:

Convexifactors $\longrightarrow$ **Multiobjective Programming** $\longrightarrow$

Bilevel Programming $\longrightarrow$ **Bilevel Multiobjective Duality**

Convexifactors. Jeyakumar and Luc [19] proposed the use of convexifactors as closed, not necessarily convex, sets that represent information about directional derivatives. The chain rule calculus was developed further by Dutta and Chandra [3], together with a mean-value theorem and the concept of ∂ -pseudoconvexity, making convexifactors a rigorous and tractable tool for nonsmooth analysis.

Multiobjective programming. Dutta and Chandra [3] used convexifactors in vector minimisation, where the classical gradient is replaced by a set-valued object. By doing so, they derived necessary conditions and showed that the resulting conditions can be sharper than those obtained from Clarke subdifferential theory.

Bilevel programming. Suneja and Kohli [7] introduced convexifactors into the scalar bilevel setting through the value-function reformulation and constructed the first duality theory in this context. Gadhi and Dempe [4] introduced the Ψ -reformulation for bilevel multiobjective problems, eliminating the constraint-qualification failure that restricts the value-function approach.

Bilevel multiobjective duality. Saini and Kailey [5] integrated all three threads—convexifactor calculus, ∂^* -convexity, and Ψ -reformulation—to obtain the first complete weak and strong duality theory for the bilevel multiobjective problem in the convexifactor setting.

7.4 Concluding Remarks

This dissertation has given a stringent and consistent treatment of duality and optimality in bilevel multiobjective programming using convexifactors, built on the work of four main papers whose results complement and strengthen one another.

Dutta and Chandra [3] provide the indispensable foundations: the convexifactor calculus, the hierarchy among different types of convexifactors, the mean-value theorem, and the conditions sufficient for weak efficient points in vector optimisation. Their key insight, that a closed non-convex set may encode directional derivative information as effectively as a convex subdifferential and may yield sharper multiplier conditions, forms the basis of the subsequent development.

This calculus was then adapted to bilevel optimisation by Suneja and Kohli [7], who developed a complete theory for the scalar bilevel case. Their introduction of five classes of ∂^* -convexity, together with the precise logical relations among them, and their construction of Wolfe-type and Mond-Weir-type duals for the scalar bilevel problem, established the template later extended to the multiobjective case.

The necessary reformulation was provided by Gadhi and Dempe [4]. They showed how the implicit bilevel feasibility condition can be converted, through the Ψ -function, into an explicit locally Lipschitz scalar inequality for which standard constraint qualifications can hold. This made it possible to derive KKT-type necessary conditions for the multiobjective bilevel problem without relying on the value-function approach.

All these threads were then brought together by Saini and Kailey [5], who delivered the first complete strong weak duality results for the bilevel multiobjective problem in the convexifactor framework. Their strong duality theorem states that any primal weak-efficient solution satisfying the non-smooth Abadie constraint qualification is associated with a dual weak-efficient solution with the same objective value, and this is the main outcome of the dissertation.

The significance of convexifactors is twofold. Analytically, they may be sharper than Clarke subdifferentials, since their convex hull can be strictly smaller, leading to tighter multiplier conditions. Structurally, their chain-rule calculus is directly compatible with the nonsmooth composite functions produced by the Ψ -reformulation, making them a natural analytical tool for the class of problems studied here.

7.5 Future Scope

The model formulated in this dissertation indicates several valuable extensions.

1. **Design of computational techniques.** The theoretical findings of this dissertation are yet to be converted into useful algorithms for solving bilevel multiobjective problems. The stationarity system of Theorem 6.1, expressed in terms of convexifactors, can offer a foundation for bundle-type or proximal-point methods in the nonsmooth bilevel setting. The construction of such techniques, the demonstration of convergence under the assumptions of ∂^* -convexity, and testing them on standard problems remain important directions for future research.
2. **Real-world applications.** Bilevel multiobjective problems arise in transportation, environmental policy, supply chain management, and machine learning. The convexifactor framework developed here provides a rigorous setting for examining such problems. The next significant step is to translate assumptions such as the non-smooth Abadie constraint qualification and ∂^* -convexity into verifiable conditions for specific applications.
3. **Mond-Weir duality for the multi-objective bilevel problem.** A Mond–Weir-type dual for the scalar bilevel problem under weaker assumptions than Wolfe duality was developed

by Suneja and Kohli [7]. A corresponding Mond-Weir dual for the multi-objective bilevel problem under the Ψ -reformulation and convexifactor framework has not yet been properly established, and this remains one of the key theoretical gaps.

4. **Stochastic and robust bilevel models.** The current framework is deterministic, whereas most real-life bilevel problems involve uncertainty. Extending the Ψ -reformulation and convexifactor duality theory to stochastic and robust bilevel models would require additional tools from stochastic and robust optimisation and remains an important open problem.
5. **Extension to hierarchical models with many levels.** This dissertation deals only with bilevel problems. Extending the reformulation and convexifactor duality theory to tri-level or k -level multiobjective programmes, together with suitable constraint qualifications and convexity assumptions, would greatly broaden the scope of the present work.

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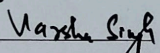
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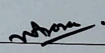
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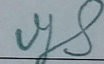
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